WEBSITE DEVELOPMENT

Table of content

Table of content 1

Acronyms 2

Learning Unit 1: Design simple web page layouts 5

Learning outcome 1.1 Identify of main layout sections according to ndesign specifications. 6

Learning outcome 1.2 Design the web page Layout to match design specifications. 12

Learning Unit 2: Develop Cascarding Style Sheets (CSS) 51

Learning outcome 2.1 Use the appropriate Format, style and layout of the elements on a web page as design specifications 52

Learning Unit 3: Ensure website content meets technical protocols and standards. 80

Learning Outcome 3.1 Verify and validate the web content in accordance with required standards. 82

Learning Outcome3.2 Test the content as per standard procedures..... 110

Learning Unit 4: Produce interactive web animations 121

Learning Outcome 4.1: Analyze website requirements in accordance with client's requirements 123

Learning Outcome 4.2 Design of web animations and responsiveness according to the design specifications 127

Learning Outcome 4.3: Publish animations 130

Appealing Features of Bootstrap 136

Learning Unit 5: Integrate website with social web technologies..... 138

Learning Outcomes: 5.1 Identify social web to be integrated with website..... 140

Learning Outcome 5.2.Develop the code and documentation of the integration requirements 145

References: 148

Acronyms

CSS: Cascading Style Sheet

HTML: Hypertext Markup Language

UI: User Interfaces

SEO: search engine optimization

APNG: Animated Portable Network Graphics

GIF: Graphics Interchange Format

ICO: Microsoft Icon

JPEG: Joint Photographic Expert Group image

PNG: Portable Network Graphics

SVG: Portable Network Graphics

Introduction

This course intends to provide basic knowledge and skills required for planning, designing and developing effective web pages with a focus on the practical application of the technologies used in the web development. It covers basic terminologies, tools, HTML, Cascading Style Sheet (CSS).

This course is intended for people who are already Software development.

The course aims to:

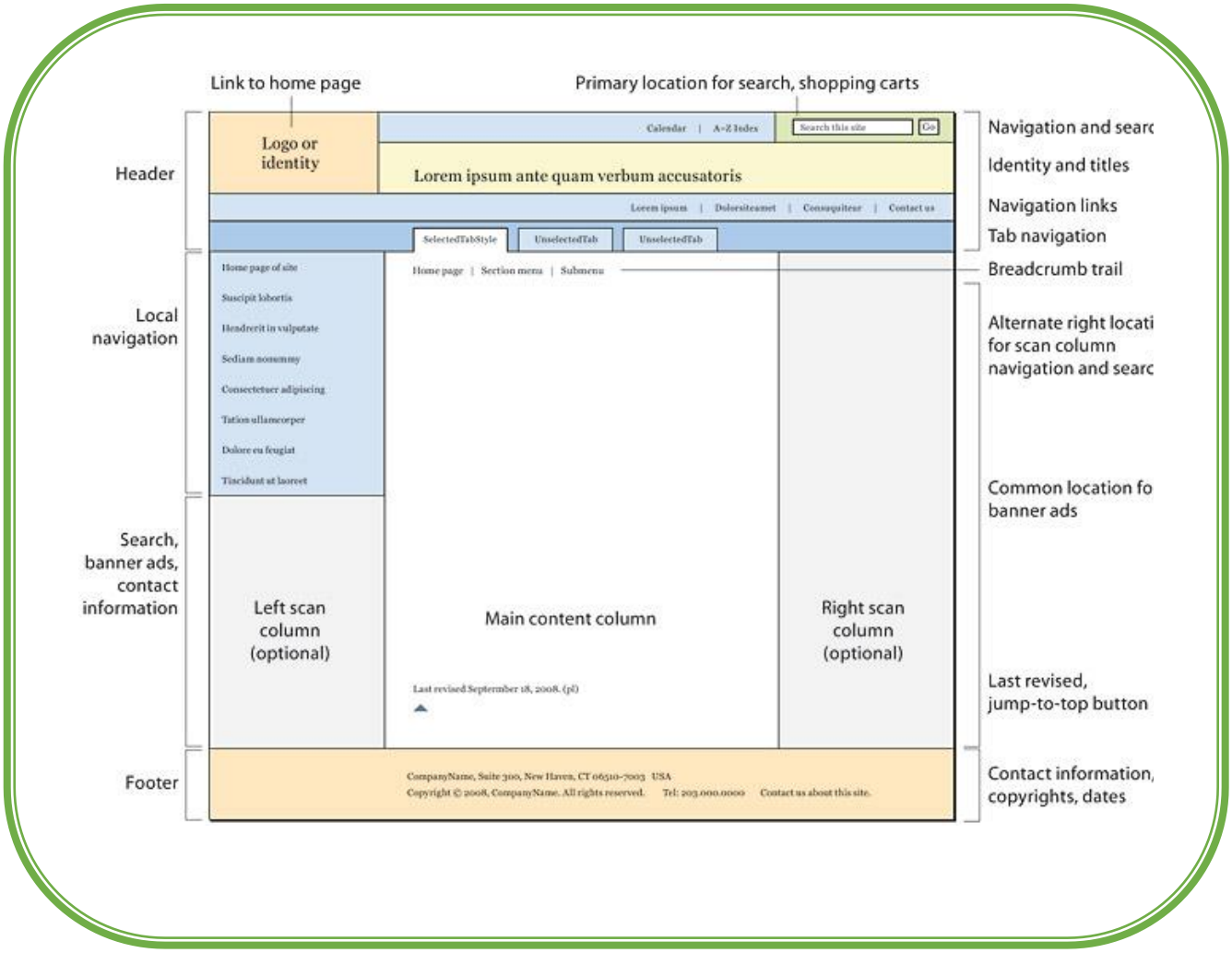
- ✓ Introduce learners the basic terminologies and various phases of website development process.
- ✓ Help learners to recognize and understand HTML web page elements and HTML tags.
- ✓ Enable learners to develop web pages using HTML.
- ✓ Make learners capable to apply CSS to format web page elements.
- ✓ Ensure website content meets technical protocols and standards
- ✓ Produce interactive web animation
- ✓ Integrate social web technologies

Module Code and Title: **SFDDW401: WEBSITE DEVELOPMENT**

Learning Units:

1. Design simple web page layouts
2. Develop cascading style sheets
3. Ensure website content meets technical protocols and standards
4. Produce interactive web animation
5. Integrate social web technologies

Learning Unit 1: Design simple web page layouts



STRUCTURE OF LEARNING UNIT

Learning outcomes:

1.1. Identify of main layout sections according to design specifications.

1.2. Design the web page Layout to match design specifications.

Learning outcome 1.1 Identify of main layout sections according to ndesign specifications.

Duration: 2hrs
Learning outcome 1.1 objectives: **By the end of Identify of main layout sections according to design specifications, the trainee will be able to:** 1. Identify basic Page structure and site design. 2. Describe the types of User Interface

Resources

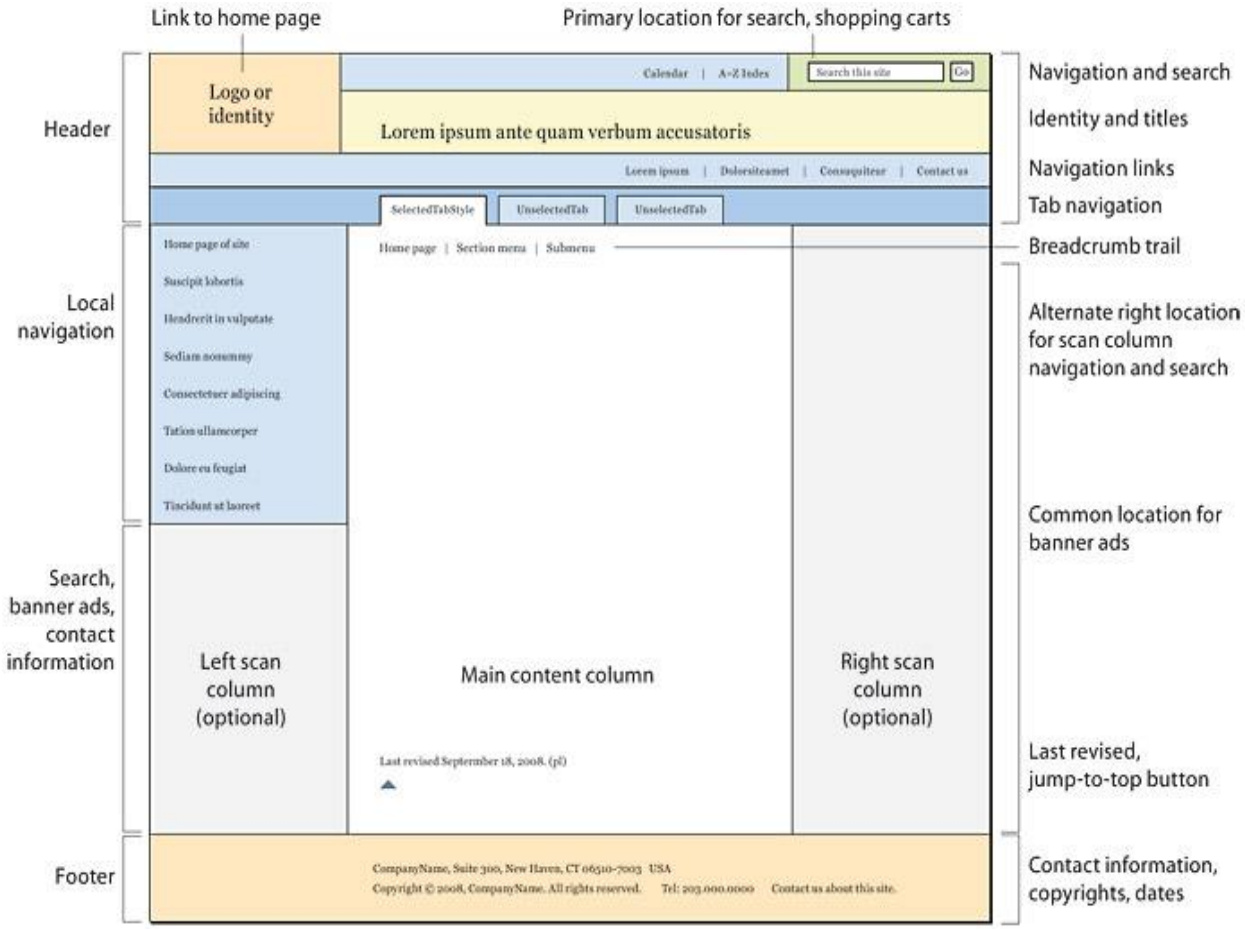
Equipment	Tools	Materials
✓ Computer.	✓ Papers. ✓ Markers. ✓ Board.	✓ Books. ✓ Pens. ✓ Pencils.

CONTENT

1.1.1 Page structure Deffinition

Page structure and **site design** are the basic parts that internet users often expect to see when they visit a web site.

Understanding the location and propose of the the main structures can help you relay information about your small business and products or services in ways that attract visitors and retain their interest.



1.1.2 Basic Page structure and site design.

A. Page headers

The header is the area that runs horizontally across the top of a page and is commonly the same on most every page in the site. It helps make a website visually identifiable to visitors. Similar to a letter heading or letterhead at the

top of stationery, the page header displays information about the person or company controlling the website via title text, log, background images, tagline or a combination of these elements. Other elements often placed in the header include a site-search box, shopping cart link, site-access link and navigation tools.

B. Navigation Tools.

Web-page navigation tools are located in several areas outside of the header including the right or left sides, center or boottom of the page. They offer page-to-page navigation or instant jump to the top of the current page. Designs feature text or image based one click links organized standalone or in tab, drop down or pop up menu and list layouts. Some sites also feature breadcrumb trails links to every page you would visit to reach the current page organized left to right on a harizontal line in the header or top center of the page in the order of your movement through the site, if you were to follow the site's organizational hierarchy.

C. Primary content

The primary content area on a page is traditionally located to the left or right of sidebar or between two sidebars. It provides main page information you want a visitor to focus on. The primary content area features a main title and content formatted into concise text paragraphs, images, videos or combination elements divided by spaces or subheadings. It also often features elements previously mentioned such as a breadcrumb trail and jump navigation links, as well as update information such as content publication or update dates and links to websites relevant to the content or that you think would interest visitors.

D. Sidebar columns.

Sidebar columns, also knowns as sidebars, run vertically along the left or right side of web pages. They usually provide primary or secondary site navigation links and information you want to emphasize such as contact details or important updates about the site operator or the topic of the site. Other elements often placed in sidebars include personal or partner advertising, a site search box and search filter tools. Sidebars usually display information as an unbroken column or a column divided into sections or boxer.

E. Page Footer

The footer runs horizontally across the bottom of pages. It provides navigation links visitors might find useful, as well as details about a page or website such as a logo, copyright date, website operator's name, page author name, legal statements and links to the site terms of use and privacy policies. Other elements aften placed in the foot include links to the site operator's contact page oe email address, job postings page, feedback form page, support page and frequently asked questions page.

1.1.3. user interfaces (UI)

1.1.3.1. The user interface Definition

The user interface is the point at which human users interact with a computer, website or application. The goal of effective UI is to make the user's experience easy and intuitive, requiring minimum effort on the user's part to receive the maximum desired outcome.

1.1.3.2. Why is user interface important?

User interface (UI) is important to meet user expectations and support the effective functionality of your site. A well-executed user interface facilitates

effective interaction between the user and the program, app or machine through contrasting visuals, clean design and responsiveness.

1.1.3.3. Types of User interface

Types of user interfaces can include:

- **Form-based user interface:** Used to enter data into a program or application by offering a limited selection of choices. For example, a settings menu on a device is form-based.
- **Graphical user interface:** A tactile UI input with a visual UI output (keyboard and monitor).
- **Menu-driven user interface:** A UI that uses a list of choices to navigate within a program or website. For example, ATMs use menu-driven UIs and are easy for anyone to use.
- **Touch user interface:** User interface through haptics or touch. Most smartphones, tablets and any device that operates using a touch screen use haptic input.
- **Voice user interface:** Interactions between humans and machines using auditory commands. Examples include virtual assistant devices, talk-to-text and GPS.

Summary for the trainer related to the content

In this unit, we learned how to Understanding the location and propose of the the main structures can help you relay information about your small business and products or services in ways that attract visitors and retain their interest.

Points to Remember (Take home message)

- ✓ Describe types of User Interface.
- ✓ Identify basic Page structure and site design

Learning outcome 1.1: formative assessment

Please mix different assessment instruments/tools for triangulation and relevancy of assessment

Written assessment

1. A user interface (UI) is the space where interactions between humans and machines occur. Describe two (2) types of User Interface.
2. Identify any five (3) basic Page structure and site design

Learning outcome 1.2 Design the web page Layout to match design specifications.

Duration: 8hrs

Learning outcome 1.2 objectives:

By the end of design, the web page Layout to match design specifications, the trainee will be able to:

1. Define the web page, website, web browser, text Editor.
2. Differentiate between tags, element and attributes.
3. Draw Table, Numbering elements, Insert Image to web page, Link Page to another Page

1.2.1. Some Definition Related to Web Page Layout

- **HTML:** is Hypertext Markup Language for creating Web pages.

Is also responsible for telling a Web browser (e.g., Microsoft Internet Explorer, Mozilla Firefox, Opera, Mac Safari, Google Chrome, and so on) how text and other objects in a Web document should appear. Whether the text should be small, large, bold, underlined, or right or left justified is largely determined by the HTML embedded in a Web page.

- **A web page:** a web page is a collection of code written in html by using

html tags, and is a document which can be displayed in a web browser such as Firefox, Google Chrome, Opera, Microsoft Internet Explorer or Edge, or Apple's Safari.

- **Website** is a collection of related network web resources, such as web pages, multimedia content, which are typically identified with a common domain name, and published on at least one web server.

Software namely, Web browsers. You create content specifying how the browser should Display it (highlighting certain pieces of text, and so on). When the browser displays the page, it applies the appropriate formatting accordingly so the user sees the text and document as you intended. You need a way to mark up the text so the browser understands it.

In the early 1990s, a Hypertext Markup Language (HTML) was created, it is not a programming language but **a language created to provide a way for users to markup documents so Web browsers could display certain elements of the document in italics, underlined, and so on.**

Several requirements must be considered when telling a computer how to format text. A short

List of the requirements includes the following:

- ✓ The instructions should follow a stringent set of guidelines.
- ✓ The instructions should be included in the textual document.
- ✓ The instructions should be invisible to the end user.

The instructions should tell the display device (usually a Web browser) where to start and end, and how to apply the formatting specified

- **A web browser** (also referred to as an **Internet browser** or simply

a **browser**) is application software for accessing the World Wide Web or a local website. When a user requests a web page from a particular website, the web

browser retrieves the necessary content from a web server and then displays the page on the user's device.

The purpose of a web browser (Chrome, Edge, Firefox, Safari) is to read HTML documents and display them correctly.

A browser does not display the HTML tags, but uses them to determine how to display the document:

Examples of browser

1. *Mozilla firefox*
2. *Google chrome*
3. *Safari*
4. *Opera*
5. *Internet Explorer*
6. *Microsoft Edge*
7. *Netscape Navigator*
8. *Brave*
9. *Vivaldi*
10. *Chromium*
11. *Tor browser*
12. *Maxthon*
13. *Yandex browser*
14. *Lynx*
15. *Epic*
16. *SeaMonkey*
17. *UC Browse*

- **An HTML editor** is a software for editing and creating HTML code that is

used for websites or other web documents. With text-based HTML editors, the source code can be edited directly.

Examples of HTML Editors

- ✓ Atom
- ✓ Notepad++
- ✓ Sublime Text
- ✓ Adobe Dreamweaver CC
- ✓ Visual Studio Code

Below is a visualization of an HTML page structure:

```
<html>
  <head>
    <title>Page title</title>
  </head>
  <body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
    <p>This is another paragraph.</p>
  </body>
</html>
```

1.2.2. HTML TAG definition

HTML TAGS are instructions used to format a web page content.

These instructions are delimited. This means that they are enclosed in angle brackets more commonly known as "**less than''** and "**greater than**" signs **(< and >)**.

The beginning marks contains a keyword corresponding to what the markup should accomplish. The ending mark includes **a slash (/).**

Tags are the starting and ending parts of an HTML element. They begin with **<** symbol and end with **>** symbol. Whatever written inside < and > are called tags.

Example:

1.2.2.1. Types of Tags in HTML

An HTML document is created using different types of tags. HTML tags can be defined and divided based on a different basis. We have divided HTML tags based on the following classifications:

A. Paired and Unpaired Tags

Following are the paired and unpaired tags in HTML explained in detail with the help of examples.

A.1 Paired Tags

An HTML tag is known as a paired tag when the tag consists of an **opening tag** and **a closing tag** as its companion tag. An HTML Paired tag starts with an opening tag: the tag name enclosed inside the angle brackets; for example, a paragraph opening tag is written as **' **'. The content follows the opening tag, which ends with an ending tag: the tag name starting with a forward slash; for example, an ending paragraph tag is written as **' **'. The first tag can be referred to as the 'Opening Tag', and the second tag can be called Closing Tag.

**Example ** This text is a paragraph. ** **

Output:

This text is a paragraph.

A.2 Unpaired Tags

An HTML tag is called an unpaired tag when the tag only has an opening tag and does not have a closing tag or a companion tag. The Unpaired HTML tag does not require a closing tag; an opening tag is sufficient in this type. Unpaired tags are sometimes also named as Standalone Tags or Singular Tags since they do not require a companion tag.

Example:

** ** This is a paragraph ** **

<hr>

** ** This is a bold and italicized text ** **

Here, the **<hr>** is the unpaired tag used to create a horizontal line. In older versions, you might see **hr** tag written as **<hr/>** instead of **<hr>.** These tags are also called **Empty Tag.**

A.3 Self-Closing Tags

Self-Closing Tags are those HTML tags that do not have a partner tag, where the first tag is the only necessary tag that is valid for the formatting. The main and important information is contained WITHIN the element as its attribute. An image tag is a classic example of a self-closing tag. Let's see it in action below:

Example:

```
<img src="a.jpg" alt="This is an alternate text">
```

1.2.3. HTML Elements

element is everything from the start tag to the end tag. They consist of some kind of structure or expression. It generally consists of a start tag, content and an end tag.

Example:

****This is the content. ****

Where, **** is the starting tag and **** is the ending tag.

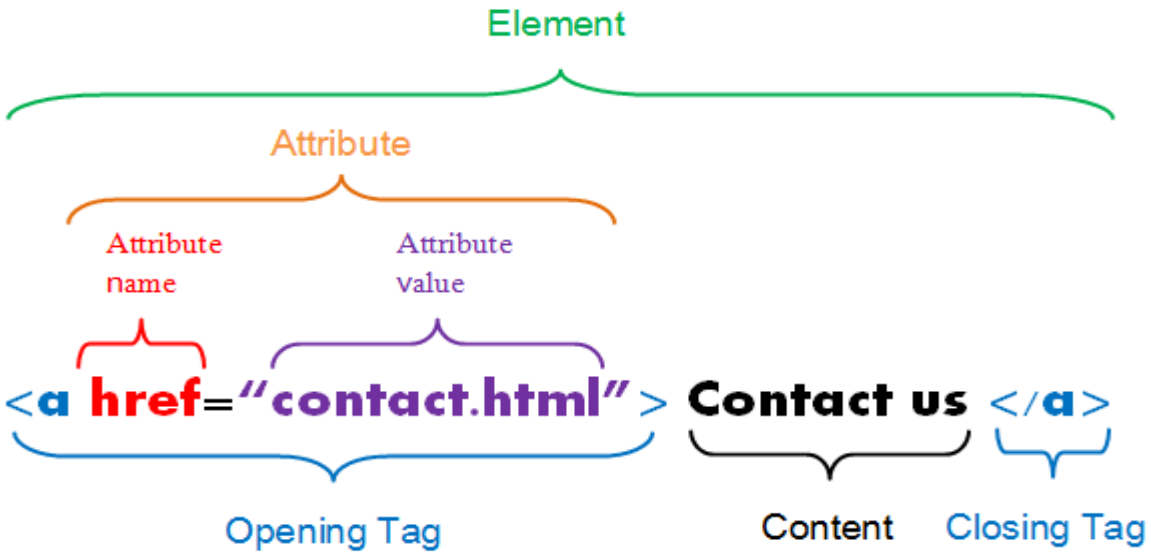
1.2.4. HTML Attributes

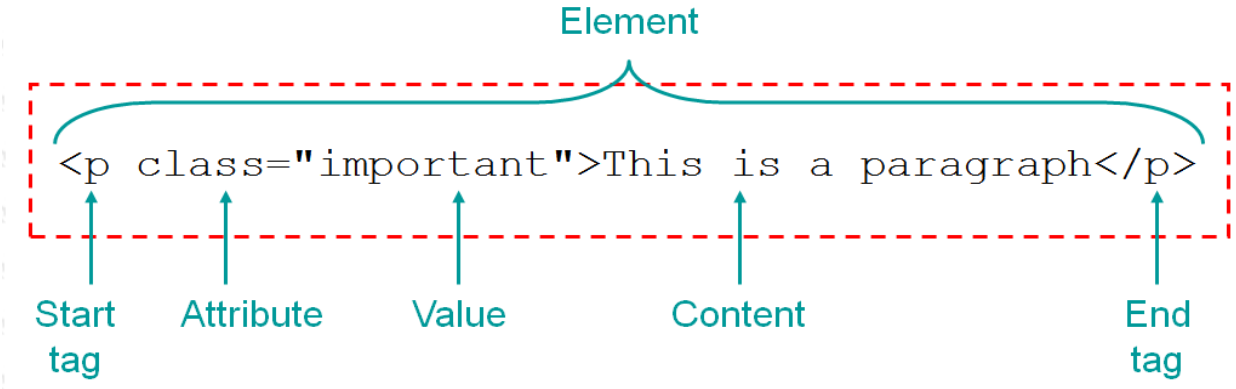
Attributes is used to define the character of an HTML element. It always placed in the opening tag of an element. It generally provides additional styling (attribute) to the element.

Example:

This is paragraph.

HTML Tags	HTML Elements	HTML Attributes
HTML tags are used to hold the HTML element.	HTML element holds the content.	HTML attributes are used to describe the characteristic of an HTML element in detail.
HTML tag starts with < and ends with >	Whatever written within a HTML tag are HTML elements?	HTML attributes are found only in the starting tag.
HTML tags are almost like keywords where every single tag has unique meaning.	HTML elements specifies the general content.	HTML attributes specify various additional properties to the existing HTML element.





1.2.5. HTML Elements Category

1.2.5.1. Basic HTML Elements

<!DOCTYPE>: Defines the document type.

All HTML documents must start with a **<!DOCTYPE>** declaration.

The declaration is not an HTML tag. It is an "information" to the browser about what document type to expect.

- **<html>**: ***Defines an HTML document***

  The **<html>** element is the root element and it defines the whole HTML document. It has a start tag **<html>** and an end tag **</html>.**

- **<body>**: ***Defines the document's body.***

  The **<body>** element contains all the contents of an HTML document, such as headings, paragraphs, images, hyperlinks, tables, lists, etc

  <head>: ***Contains metadata/information for the document.***

  Contains metadata/information for the document.

  The **<head>** element is a container for **metadata** (data about data) and is placed between the **<html>** tag and the **<body>** tag.

  Metadata is data about the HTML document. Metadata is not displayed.

  Metadata typically define the document title, character set, styles, scripts, and other meta information.

  The following elements can go inside the **<head>** element:

  - <title> (required in every HTML document)
  - <style>
  - <base>
  - <link>
  - <meta>
  - <script>
  - <noscript>

- **<title>: Defines a title for the document.**

The contents of a page title are very important for search engine optimization (SEO)! The page title is used by search engine algorithms to decide the order when listing pages in search results.

The **<title>** element:

- defines a title in the browser toolbar
- provides a title for the page when it is added to favorites
- displays a title for the page in search-engine results

- **<h1> to <h6>: Defines HTML headings.**

<h1> defines the most important heading. **<h6>** defines the least important heading.

Example

The six different HTML headings:

```
<h1>This is heading 1</h1>
<h2>This is heading 2</h2>
<h3>This is heading 3</h3>
<h4>This is heading 4</h4>
<h5>This is heading 5</h5>
<h6>This is heading 6</h6>
```

Output:

This is heading 1

This is heading 2

This is heading 3

This is heading 4

This is heading 5

This is heading 6

- **<p>: Defines a paragraph.**

  Browsers automatically add a single blank line before and after each **<p>** element.

- **
: Inserts a single line break.**

  Inserts a single line break.

  The **
** tag is useful for writing addresses or poems.

Example

Use <br> in a poem:

```
<p>Be not afraid of greatness.<br>
Some are born great,<br>
some achieve greatness,<br>
and others have greatness thrust upon them.</p>
<p><em>-William Shakespeare</em></p>
```

Output:

A Poem

Be not afraid of greatness.
Some are born great,
some achieve greatness,
and others have greatness thrust upon them.

-William Shakespeare

The <br> tag is an empty tag which means that it has no end tag.

- **<hr>: Defines a thematic change in the content.**

The **<hr>** tag defines a thematic break in an HTML page.

The **<hr>** element is most often displayed as a horizontal rule that is used to separate content (or define a change) in an HTML page.

Examples of difference between <br> and <hr>

```
<h1>The Main Languages of the Web</h1>

<p>HTML is the standard markup language for creating Web pages. HTML describes the structure of a Web page, and consists of a series of elements. HTML elements tell the browser how to display the content. </p>
<hr>
<p>CSS is a language that describes how HTML elements are to be displayed on screen, paper, or in other media. CSS saves a lot of work, because it can control the layout of multiple web pages all at once. </p>
<hr>
<p>JavaScript is the programming language of HTML and the Web. JavaScript can change HTML content and attribute values.
```

JavaScript can change CSS. JavaScript can hide and show HTML elements, and more. </p>

Output:

The Main Languages of the Web

HTML is the standard markup language for creating Web pages. HTML describes the structure of a Web page, and consists of a series of elements. HTML elements tell the browser how to display the content.

CSS is a language that describes how HTML elements are to be displayed on screen, paper, or in other media. CSS saves a lot of work, because it can control the layout of multiple web pages all at once.

JavaScript is the programming language of HTML and the Web. JavaScript can change HTML content and attribute values. JavaScript can change CSS. JavaScript can hide and show HTML elements, and more.

- **<! --...-->: Defines a comment.**

The comment tag is used to insert comments in the source code. Comments are not displayed in the browsers. You can use comments to explain your code, which can help you when you edit the source code at a later date. This is especially useful if you have a lot of code.

Example

```
<! --This is a comment. Comments are not displayed in the browser-->
<p>This is a paragraph. </p>
```

1.2.5.2. HTML Formatting Elements

Formatting elements were designed to display special types of text:

- **<b>**: Bold text

The HTML **** element defines bold text, without any extra importance.

Example:

****This text is bold****

Output:

This text is bold

- **<u>**: underline text

The **<u>** tag in HTML stands for underline, and it's used to underline the text enclosed within the **<u>** tag. This tag is generally used to underline misspelled words. This tag requires a starting as well as ending tag.

Syntax:

<u> Contents… </u>

Example:

<u>This text is bold**</u>**

Output:

<u>This text is bold</u>

- ****: Important text.

The HTML **** element defines text with strong importance. The content inside is typically displayed in bold.

Example:

****This text is important**! **

Output:

This text is important!

- **:** Italic text.

  The HTML **** element defines a part of text in an alternate voice or mood. The content inside is typically displayed in italic.

  Example:

  ****This text is italic****

  Output:

  This text is italic

- **:** Emphasized text

  The HTML **** element defines emphasized text. The content inside is typically displayed in italic.

  Example:

  ****This text is emphasized****

  Output:

  This text is emphasized

- **<mark>:** Marked text.

  The HTML **<mark>** element defines text that should be marked or highlighted

  Example:

  Do not forget to buy **<mark>**milk**</mark>** today**.**

  Output: Do not forget to buy milk today.

- **<small>:** Smaller text.

  The HTML **<small>** element defines smaller text:

  Example:

  <small>This is some smaller text**. </small>**

  Output: This is some smaller text.

- **<del>:** Deleted text

  The HTML **<del>** element defines text that has been deleted from a document. Browsers will usually strike a line through deleted text

  Example:
  My favorite color is **<del>**blue**</del>** red.
  Output: My favorite color is ~~blue~~ red.

- **<ins>:** Inserted text.

  The HTML **<ins>** element defines a text that has been inserted into a document. Browsers will usually underline inserted text:

  Example:
  My favorite color is <del>blue</del> **<ins>**red**</ins>**
  Output: My favorite color is ~~blue~~ red

- **<sub>:** Subscript text.

The HTML **<sub>** element defines subscript text. Subscript text appears half a character below the normal line, and is sometimes rendered in a smaller font. Subscript text can be used for chemical formulas, like H_2O:

Example:

This is **_{**subscripted**}** text.

<p>

H**_{**2**}**O

Output:

This is $_{\text{subscripted}}$ text.

H_2O

- **<sup>:** Superscript text.

The HTML <sup> element defines superscript text. Superscript text appears half a character above the normal line, and is sometimes rendered in a smaller font. Superscript text can be used for footnotes.

Example:

This is ^{superscripted} text.

<p>

125**^{**2**}**

Output:

This is ^{superscripted} text.

125^2

1.2.5.3. HTML **<img>** *tag*

HTML ***<img>*** tag is used to add image inside webpage/website. Nowadays website does not directly add images to a web page, as the images are linked to web pages by using the **<img>** tag which holds space for the image.

Syntax:

<img src="" alt="" width="" height="">

Attributes: The ***<img>*** tag has following attributes.

- **src:** It is used to specify the path to the image.
- **alt:** It is used to specify an alternate text for the image. It is useful as it informs the user about what the image means and also due to any network issue if the image cannot be displayed then this alternate text will be displayed.
- **crossorigin:** It is used to import images from third-party sites that allow cross-origin access to be used with canvas.
- **height:** It is used to specify the height of the image.
- **width:** It is used to specify the width of the image.
- **ismap:** It is used to specify an image as a server-side image map.
- **loading:** It is used to specify whether a browser should defer loading of images until some conditions are met or load an image immediately.
- **longdesc:** It is used to specify a URL to a detailed description of an image.
- **referrerpolicy**: It is used to specify which referrer information to use when fetching an image i.e. *no-referrer, no-referrer-when-downgrade, origin, origin-when-cross-origin, unsafe-url.*
- **sizes:** It is used to specify image sizes for different page layouts.
- **srcset:** It is used to specify a list of image files to use in different situations.
- **usemap:** It is used to specify an image as a client-side image map.

Example1: In this example we are using the <img> tag along with ***src***, *width*, *height*, and *alt* attributes.

```
<html>
<body style="text-align: center;">
  <h3>GeeksforGeeks logo</h3>
    <img src=
   "https://media.geeksforgeeks.org/wp-content/uploads/geeksforgeeks-
                                                    13.png"
    width="420" height="100"
    alt="Geeksforgeeks.org">
```

```
</body>
</html>
```

Output:

img src with width and height

Here are the most common image file types, which are supported in all browsers (Chrome, Edge, Firefox, Safari, Opera):

Abbreviation	File Format	File Extension
APNG	Animated Portable Network Graphics	.apng
GIF	Graphics Interchange Format	.gif
ICO	Microsoft Icon	.ico, .cur
JPEG	Joint Photographic Expert Group image	.jpg, .jpeg, .jfif, .pjpeg, .pjp
PNG	Portable Network Graphics	.png
SVG	Scalable Vector Graphics	.svg

1.2.5.4. HTML Elements Lists

HTML Lists are used to specify lists of information. All lists may contain one or more list elements. There are three different types of HTML lists:

1. **Ordered List or Numbered List <ol>:** Used to create a list of related items, in no particular order.
2. **Unordered List or Bulleted List <ul>:** Used to create a list of related items, in a specific order.
3. **Description List or Definition List <dl>:** Used to create a list of terms and their descriptions.

HTML Unordered Lists

An unordered list is a collection of related items that have no special order or sequence. This list is created by using HTML **<ul>** tag. Each item in the list is marked with a bullet.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML unordered List</title>
  </head>
  <body>
    <ol>
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ol>
  </body>
</html>
```

This will produce the following result

- Beetroot
- Ginger
- Potato
- Radish

You can use **type** attribute for **<ul>** tag to specify the type of bullet you like. By default, it is a **disc**. Following are the possible options

1. <ul type = "square">
2. <ul type = "disc">
3. <ul type = "circle">

Example for each type attribute

1. **Example where we used <ul type = "square">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Unordered List</title>
  </head>
  <body>
    <ul type = "square">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body
</html>
```

This will produce the following result

- Beetroot
- Ginger
- Potato
- Radish

2. Example where we used <**ul type = " disc**">

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Unordered List</title>
  </head>
  <body>
```

This will produce the following result

- Beetroot
- Ginger
- Potato

```
    <ul type = "disc">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body>
</html>
```

- Radish

3. Example where we used **<ul type = " circle">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Unordered List</title>
  </head>
  <body>
    <ul type = "circle">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body>
</html>
```

This will produce the following result

- Beetroot
- Ginger
- Potato
- Radish

HTML Ordered Lists

Ordered Lists are used to put items in a numbered list instead of bulleted. This list is created by using **<ol>** tag. The numbering starts at one and is incremented by one for each successive ordered list element tagged with **<li>.**

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Ordered List</title>
  </head>
<body>
    <ol>
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ol>
  </body>
</html>
```

This will produce the following result

1. Beetroot
2. Ginger
3. Potato
4. Radish

Ordered List use different type attribute for <ol> tag to specify the type of numbering you like. By default, it is a number.

Following are the possible option

- ✓ **<ol type = "1">** Default-Case Numerals.
- ✓ **<ol type = "I">** Upper-Case Numerals.
- ✓ **<ol type = "i">** Lower-Case Numerals.
- ✓ **<ol type = "A">** Upper-Case Letters.
- ✓ **<ol type = "a">** Lower-Case Letters.

1. Example where we used **<ol type = "1">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML ordered List</title>
  </head>
  <body>
    <ul type = "1">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
```

This will produce the following result

1. Beetroot
2. Ginger
3. Potato
4. Radish

```
      <li>Radish</li>
    </ul>
  </body>
</html>
```

2. Example where we used **<ol type = " I ">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML ordered List</title>
  </head>
  <body>
    <ul type = " I ">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body>
</html>
```

This will produce the following result

I. Beetroot
II. Ginger
III. Potato
IV. Radish

3. Example where we used **<ol type = " i ">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML ordered List</title>
  </head>
  <body>
    <ul type = " i ">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
```

This will produce the following result

i. Beetroot
ii. Ginger
iii. Potato
iv. Radish

```
    </ul>
  </body>
</html>
```

4. Example where we used **<ol type = " A ">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML ordered List</title>
  </head>
  <body>
    <ul type = " A ">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body>
</html>
```

This will produce the following result

A. Beetroot
B. Ginger
C. Potato
D. Radish

5. Example where we used **<ol type = " a ">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML ordered List</title>
  </head>
  <body>
    <ul type = " a ">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body></html>
```

This will produce the following result

a. Beetroot
b. Ginger
c. Potato
d. Radish

Ordered List use also different **start** attribute for **<ol>** tag to specify the starting point of numbering you need.

Possible options used "start attribute"

- ✓ **<ol type = "1" start = "4">** Numerals starts with 4.
- ✓ **<ol type = "I" start = "4">** Numerals starts with IV.
- ✓ **<ol type = "i" start = "4">** Numerals starts with iv.
- ✓ **<ol type = "a" start = "4">** Letters starts with d.
- ✓ **<ol type = "A" start = "4">** Letters starts with D.

Example where we used **<ol type = " i " start = "4">**

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML ordered List</title>
  </head>
  <body>
    <ul type = " i " start = "4">
      <li>Beetroot</li>
      <li>Ginger</li>
      <li>Potato</li>
      <li>Radish</li>
    </ul>
  </body>
</html>
```

This will produce the following result

iv. Beetroot
v. Ginger
vi. Potato
vii. Radish

Description List or Definition List <dl>

Definition list <dl>: is used to list items along with a description of each item. Definition lists are typically formatted with the term on the left with the definition following on the right or on the next line. The **<dl>** tag is used in conjunction with **<dt>** (defines terms/names) and <dd> (describes each term/name).

Defines a term/name <dt>

The **<dt>** tag defines a term/name in a description list. The **<dt>** tag is used in conjunction with **<dl>** (defines a description list) and **<dd>** (describes each term/name).

Defines a description <dd>

The **<dd>** tag is used to describe a term/name in a description list. The **<dd>** tag is used in conjunction with <dl> (defines a description list) and <dt> (defines terms/names). Inside a **<dd>** tag you can put paragraphs, line breaks, images, links, lists, etc.

Example of a description list, with terms and descriptions:

<!DOCTYPE html> <html> <body> <h3>The dl, dd, and dt elements</h3> These three elements are used to create a description list: <dl> <dt>Coffee</dt> <dd>Black hot drink</dd> <dt>Milk</dt> <dd>White cold drink</dd> </dl> </body> </html>	**Output:** **The dl, dd, and dt elements** These three elements are used to create a description list: Coffee Black hot drink Milk White cold drink

Summary Description of HTML Elements Lists

<ul>: Defines an unordered list

<ol>: Defines an ordered list

<li>: Defines a list item

<dl>: Defines a description list

<dt>: Defines a term/name in a description list

<dd>: Defines a description of a term/name in a description list

1.2.5.5. The HTML tables Elements

The HTML tables allow web authors to arrange data like text, images, links, other tables, etc. into rows and columns of cells.

The HTML tables are created using the **<table>** tag in which the **<tr>** tag is used to create table rows and **<td>** tag is used to create data cells. The elements under **<td>** are regular and left aligned by default

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Tables</title>
  </head>
  <body>
    <table border = "1">
      <tr>
        <td>Row 1, Column 1</td>
        <td>Row 1, Column 2</td>
      </tr>
      <tr>
        <td>Row 2, Column 1</td>
        <td>Row 2, Column 2</td>
      </tr>
    </table>
  </body>
</html>
```

Output:

Row 1, Column 1	Row 1, Column 2
Row 2, Column 1	Row 2, Column 2

Here, the **border** is an attribute of <table> tag and it is used to put a border across all the cells. If you do not need a border, then you can use border = "0".

A. Table Heading

Table heading can be defined using **<th>** tag. This tag will be put to replace **<td>** tag, which is used to represent actual data cell. Normally you will put your top row as table heading as shown below, otherwise you can use **<th>** element in any row. Headings, which are defined in **<th>** tag are centered and bold by default

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table Header</title>
  </head>
  <body>
    <table border = "1">
      <tr>
        <th>Name</th>
        <th>Salary</th>
      </tr>
      <tr>
        <td>Ramesh Raman</td>
        <td>5000</td>
      </tr>
      <tr>
        <td>Shabbir Hussein</td>
        <td>7000</td>
      </tr>
    </table>
  </body>
</html>
```

Output:

Name	Salary
Ramesh Raman	5000
Shabbir Hussein	7000

B. Cellpadding and Cellspacing Attributes

There are two attributes called ***cellpadding*** and ***cellspacing*** which you will use to adjust the white space in your table cells.

The cellspacing attribute defines space between table cells, while

cellpadding represents the distance between cell borders and the content within a cell.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table Cellpadding</title>
  </head<body>
    <table border = "1" cellpadding = "5"
cellspacing = "5">
      <tr>
        <th>Name</th>
        <th>Salary</th>
      </tr> <tr>
        <td>Ramesh Raman</td>
        <td>5000</td>
      </tr><tr>
        <td>Shabbir Hussein</td>
        <td>7000</td> </tr> </table>
  </body></html>
```

Output:

Name	**Salary**
Ramesh Raman	5000
Shabbir Hussein	7000

C. Colspan and Rowspan Attributes

You will use **colspan** attribute if you want to merge two or more columns into a single column. Similar way you will use **rowspan** if you want to merge two or more rows.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table
Colspan/Rowspan</title>
  </head>
  <body>
    <table border = "1">
      <tr>
        <th>Column 1</th>
        <th>Column 2</th>
        <th>Column 3</th>
      </tr>
      <tr>
        <td rowspan = "2">Row 1 Cell
1</td>
        <td>Row 1 Cell 2</td>
        <td>Row 1 Cell 3</td>
      </tr>
      <tr>
        <td>Row 2 Cell 2</td>
        <td>Row 2 Cell 3</td>
      </tr>
      <tr>
        <td colspan = "3">Row 3 Cell
1</td>
      </tr>
    </table>
  </body>
</html>
```

Output:

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

D. Tables Backgrounds

You can set table background using one of the following two ways:

- **bgcolor** attribute: You can set background color for whole table or just for one cell. You can also set border color also using **bordercolor** attribute.
- **background** attribute: You can set background image for whole table or just for one cell.

Example Using bgcolor or bordercolor

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table
Background</title>
  </head>
  <body>
    <table border = "1" bordercolor =
"green" bgcolor = "yellow">
      <tr>
        <th>Column 1</th>
        <th>Column 2</th>
        <th>Column 3</th>
      </tr>
      <tr>
        <td rowspan = "2">Row 1 Cell
1</td>
        <td>Row 1 Cell 2</td>
        <td>Row 1 Cell 3</td>
      </tr>
      <tr>
        <td>Row 2 Cell 2</td>
        <td>Row 2 Cell 3</td>
      </tr>
      <tr>
        <td colspan = "3">Row 3 Cell
1</td>
      </tr </table>
  </body>
</html>
```

Output:

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

Example of using **background** attribute. Here we will use an image available in /images directory.

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table
Background</title>
  </head>
  <body>
    <table border = "1" bordercolor =
"green" background = "/images/test.png">
      <tr>
        <th>Column 1</th>
        <th>Column 2</th>
        <th>Column 3</th>
```

Output:

Column 1	Column 2	Column 3
Row 1 Cell 1	Row 1 Cell 2	Row 1 Cell 3
	Row 2 Cell 2	Row 2 Cell 3
Row 3 Cell 1		

```
      </tr>
      <tr>
        <td rowspan = "2">Row 1 Cell
1</td>
        <td>Row 1 Cell 2</td><td>Row 1
Cell 3</td>
      </tr>
      <tr>
        <td>Row 2 Cell 2</td>
        <td>Row 2 Cell 3</td>
      </tr>
      <tr>
        <td colspan = "3">Row 3 Cell
1</td>
      </tr>
    </table>
  </body>
</html>
```

N.B: This will produce the following result. Here background image did not apply to table's header.

E. Table Height and Width

You can set a table width and height using **width** and **height** attributes. You can specify table width or height in terms of pixels or in terms of percentage of available screen area.

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table
Width/Height</title>
  </head>

  <body>
    <table border = "1" width = "400"
height = "150">
      <tr>
        <td>Row 1, Column 1</td>
        <td>Row 1, Column 2</td>
      </tr>
      <tr>
        <td>Row 2, Column 1</td>
        <td>Row 2, Column 2</td>
      </tr>
    </table>
  </body>
</html>
```

Output:

Row 1, Column 1	Row 1, Column 2
Row 2, Column 1	Row 2, Column 2

F. Table Header, Body, and Footer

Tables can be divided into three portions:

- **header,**
- **body, and**
- **foot.**

The head and **foot** are rather similar to headers and footers in a word-processed document that remain the same for every page, while the **body** is the main content holder of the table.

The three elements for separating the head, body, and foot of a table are:

- **<thead>**: to create a separate table header.
- **<tbody>**: to indicate the main body of the table.
- **<tfoot>** :to create a separate table footer.

A table may contain several <tbody> elements to indicate *different pages* or groups of data. But it is notable that <thead> and <tfoot> tags should appear before <tbody>

Example

```
<!DOCTYPE html>
<html>
  <head>
    <title>HTML Table</title>
  </head>
  <body>
    <table border = "1" width = "400">
      <thead>
        <tr>
          <td colspan = "4">This is the
head of the table</td>
        </tr>
      </thead>
      <tfoot>
        <tr>
          <td colspan = "4">This is the
foot of the table</td>
        </tr>
      </tfoot>
      <tbody>
        <tr>
          <td>Cell 1</td>
          <td>Cell 2</td>
          <td>Cell 3</td>
          <td>Cell 4</td>
</tr></tbody></table></body></html>
```

Output:

This is the head of the table			
Cell 1	Cell 2	Cell 3	Cell 4
This is the foot of the table			

Summary for the trainer related to the content (key notes using bullets such as ticks etc)

In this unit, we learned how to use a text editor (such as notepad) to type and save HTML code and view its output in a browser. We have also learned how to use basic HTML tags with the help of examples

Theoretical learning Activity

- ✓ What should be the extension of HTML document?
- ✓ Which two tags should be written at the beginning of an HTML document?
- ✓ Discuss basic HTML tags for converting a text into bold, italic and underlined.
- ✓ Explain the purpose of content-based tags? Discuss some of the common. content based tags. 6. Differentiate between ordered and unordered lists?

Practical learning Activity

- ✓ Write HTML code to produce the following Output.

Quick	brown fox		jumps
over the	lazy	dog	and
	then	it	fall
	prey to a hunter		

Points to Remember (Take home message)

HTML tags are used to hold the HTML element.

- ✓ HTML element holds the content.
- ✓ HTML attributes are used to describe the characteristic of an HTML element in detail.
  - ✓ <table>: Defines a table
  - ✓ <th>: Defines a header cell in a table
  - ✓ <tr>: Defines a row in a table
  - ✓ <td>: Defines a cell in a table
  - ✓ <thead>: Groups the header content in a table
  - ✓ <tbody>: Groups the body content in a table
  - ✓ <tfoot>: Groups the footer content in a table.
  - ✓ <col>: Specifies column properties for each column within a <colgroup> element

Learning outcome 1.2: formative assessment

Please mix different assessment instruments/tools for triangulation and relevancy of assessment

Written assessment

Choose the correct answer.

1. What does HTML stand for?

A. Hyper Type Makeup Longing

B. Hyper Text Markup Language

C. Hyper Type Marking Logo

D. Hypermedia Type Markup Language

2. What are the two things you need to create and view web pages?

A. A text editor and a compiler

B. A text editor and a web browser

C. A compiler and a web browser

D. A compiler and interpreter

3. Should <! DOCTYPE> declaration be the first thing in an HTML document?

A. Yes, always

B. Yes, but only when the Strict version is used

C. No

D. Both A and B

4. Which tag is the root tag in HTML?

A. <body>

B. <html>

C. <title>

D. <head>

5. Where is the title text displayed?

A. In the body of the page

B. In the status bar

C. In the title bar

D. All of the above

6. What are the different levels of headings in HTML?

A. 1-6

B. 2-5

C. 1-4

D. 1-3

7.What happens if you forget a slash at the end of a header tag?

A. A Netscape errors

B. Everything following will be a heading format

C. Nothing

D. It hides the text

8. Which tag is used to define items in an ordered or unordered list?

A. list tag

B. ls tag

C. li tag

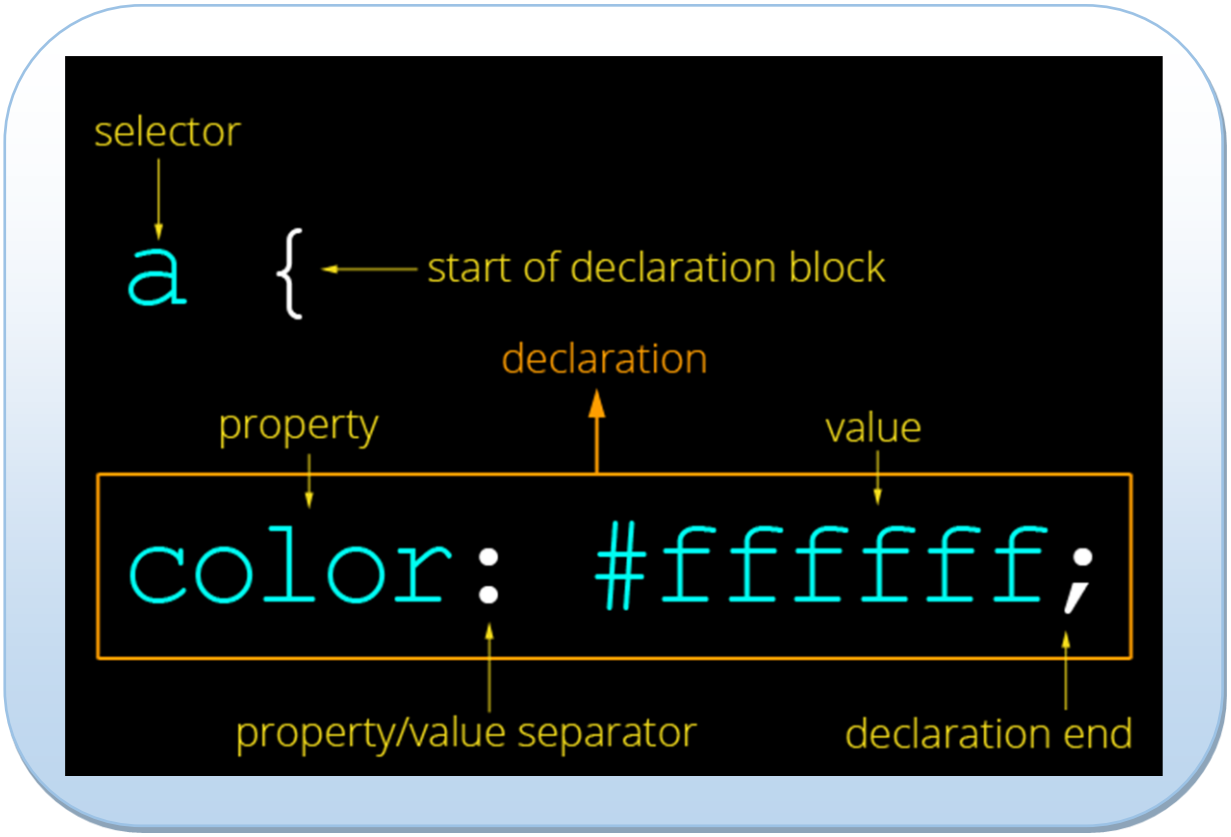
D. ol tag

Practical assessment

1.Using ordered and unordered lists, create a web page that displays months of a year grouped according to the number of days each month has. That is, all the months having 31 days should be grouped together followed by months having 30 days and February should be placed in a separate group

2. Create a web page that gives basic information about yourself. The page should display your personal information, academic qualification, professional qualification, job experience etc. Your page should be well-designed and should look attractive.

3. Create a website that provides basic information about a number of historical places in your country. Information about each place should be properly divided into sections (with proper section headings) and paragraph.



STRUCTURE OF LEARNING UNIT

Learning outcomes:
2.1 Use the appropriate Format, style and layout of the elements on a web page as design specifications.
2.2. Test and validate the layout with reference to design specifications.

Duration: 2hrs

Learning outcome 2.1 objectives:

By the end of Use the appropriate Format, style and layout of the elements on a web page as design specifications, the trainee will be able to:

- Use the HTML **style** attribute for inline styling
- Use the HTML **<style>** element to define internal CSS
- Use the HTML **<link>** element to refer to an external CSS file
- Use the HTML **<head>** element to store **<style>** and **<link>** elements
- Use the CSS **color** property for text colors
- Use the CSS **font-family** property for text fonts
- Use the CSS **font-size** property for text sizes
- Use the CSS **border** property for borders
- Use the CSS **padding** property for space inside the border
- Use the CSS **margin** property for space outside the border

Resources

Equipment	Tools	Materials
✓ Computer.	✓ Papers. ✓ Markers. ✓ Board.	✓ Books. ✓ Pens. ✓ Pencils.

Learning outcome 2.1 Use the appropriate Format, style and layout of the elements on a web page as design specifications

2.1.1 What is CSS?

Cascading Style Sheets (CSS) is used to format the layout of a webpage.

With **CSS**, you can control the color, font, the size of text, the spacing between elements, how elements are positioned and laid out, what background images or background colors are to be used, different displays for different devices and screen sizes, and much more!

The word **cascading** means that a style applied to a parent element will also apply to all children elements within the parent. So, if you set the color of the body text to "blue", all headings, paragraphs, and other text elements within the body will also get the same color.

2.1.2 What is a style sheet?

- A style sheet is a numeric document which can specify all characteristics of formatting a document linked with a tag on which it is applied.
- Styles define **how to display** HTML elements

With CSS you can define for each HTML tag, a specific format which will applied on it.

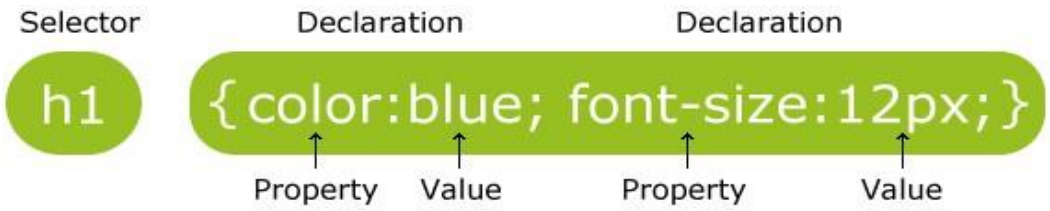
2.1.3 Utility and advantage of the style sheets

- Separate the content and formatting.
- Presentation Cohesion within the web site with the external style sheets.
- Modify the page aspect or website without changing its content in few lines instead of changing a lot of tags.
- A new language, clear, simple and logical compared to HTML with its different versions.
- Short way of writing and neat compared to HTML (reduced file size)
- It reduces the time of loading pages
- It encompasses weaknesses of HTML language (control of fonts, control of distance between lines, control of margins and indents without using

- table or <DD> tag therefore increasing creativity for those who design web sites.
- Reusability of code

CSS Syntax

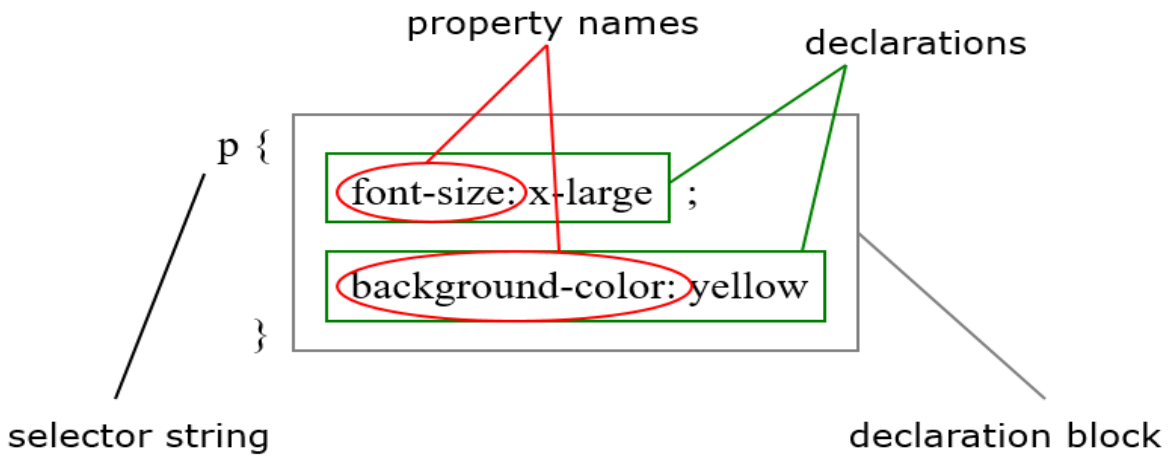
We have used CSS code without discussing how to write it. It is time to get into CSS. The purpose of a style sheet is to create a presentation for a particular HTML element or set of elements. Binding an element to a style specification is very simple. It consists of a **rule-set**. A rule-set has a **selector** and **one or more declarations**. Each declaration consists of a **property** and its **value**. The figure below sums it up:

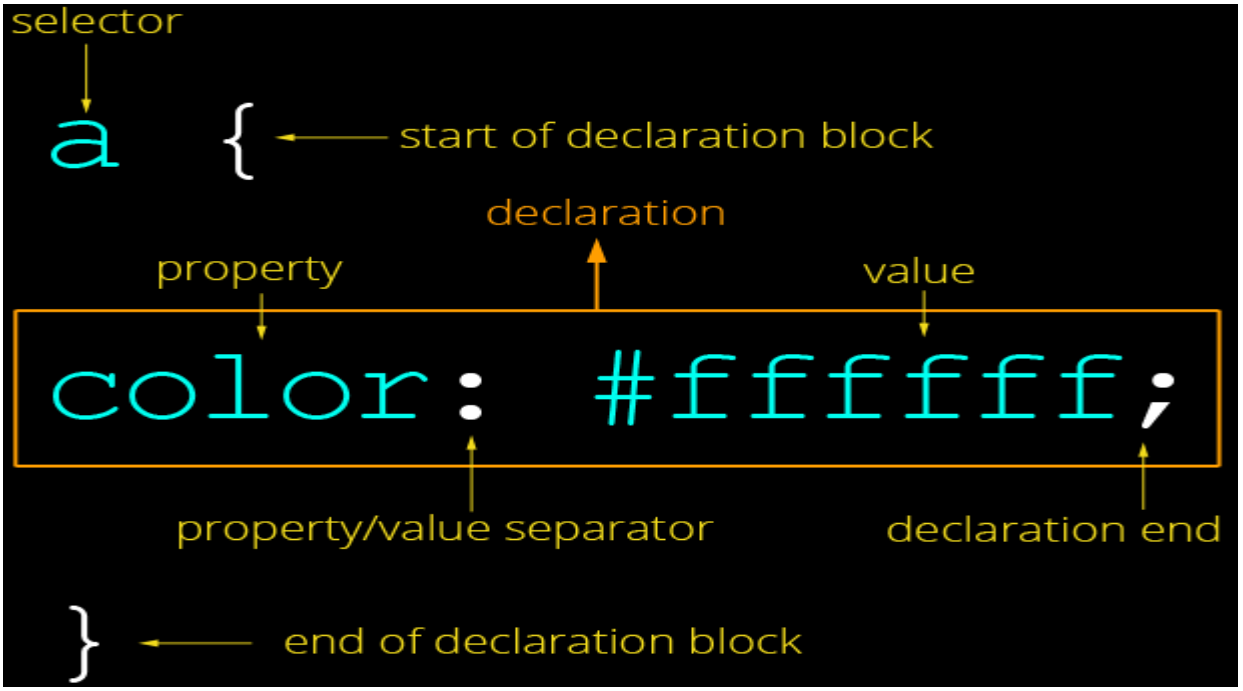


The selector refers to the HTML tag or element that we want to apply the style on. A selector can be an HTML element name, id, class, or an attribute etc. For example, the above figure defines properties for <h1> tag. A declaration is a combination of a property name and a value separated by a colon. Each declaration must end with a semi-colon, even if there is only one declaration. The declaration set must be included between curly-braces. A CSS rule-set can be written in a single line, or multiple lines. That is we can define the style for <h1> as

```
h1

{ font-size: 20px; color: red;  text-align: center;

 }
```

CSS Style Rule





A CSS rule has two main parts:

- **a selector,**
- **and one or more declarations:**

The selector is normally the HTML element you want to style.

Each **declaration** consists **of a property** and a value.

The property is the style attribute you want to change.

Properties indicate the aspects of the element you want to change. For example, color, font, width, height and border.

Each property has a value.

Values specify the settings you want to use for the chosen properties. For example, if you want to specify a color property then the value is the color you want the text in these elements to be.

2.1.5. Methods of Writing Style Sheets (CSS) CSS

CSS can be added to HTML documents in 3 ways:

A. **Inline** by using the **style attribute** inside HTML elements
B. **Internal** by using a **<style>** element in the **<head>** section
C. **External** by using a **<link>** element to link to an external CSS file

The most common way to add CSS, is to keep the styles in external CSS files. However, in this tutorial we will use inline and internal styles, because this is easier to demonstrate, and easier for you to try it yourself.

A. Inline Cascading Style Sheets

An inline CSS is used to apply a unique style to a single HTML element. An inline CSS uses the **style** attribute of an HTML element.

The following example sets the text color of the **<h1>** element to blue, and the text color of the **<p>** element to red:

Example

```
<h1 style="color:blue;">A Blue Heading</h1>

<p style="color:red;">A red paragraph.</p>
```

B. Internal Cascading Style Sheets

An internal CSS is used to define a style for a single HTML page. An internal CSS is defined in the **<head>** section of an HTML page, within a **<style>** element.

The following example sets the text color of ALL the **<h1>** elements to blue, and the text color of ALL the **<p>** elements to red. In addition, the page will be displayed with a "**powderblue**" background color:

Example

```
<!DOCTYPE html>
<html>
<head>
<style>
body {background-color: powderblue;}
h1   {color: blue;}
p    {color: red;}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>
</body>
</html>
```

C. External Cascading Style Sheets

An external style sheet is used to define the style for many HTML pages.

To use an external style sheet, add a link to it in the <head> section of each HTML page:

Example

```html
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="styles.css">
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph. </p>
</body>
</html>
```

The external style sheet can be written in any text editor. The file must not contain any HTML code, and must be saved with a .css extension.

Here is what the "styles.css" file looks like:

"styles.css":

```css
body {
  background-color: powderblue;
}
h1 {
  color: blue;
}
p {
  color: red;
}
```

C.1 Element used to insert External Cascading Style Sheets in HTML Document

<link>

The **<link>**element can be used in an HTML document to tell the browser where to find the CSS file used to style the page. It is an empty element (meaning it

does not need a closing tag), and it lives inside the element. It should use three attributes:

href

This specifies the path to the CSS file (which is often placed in a folder called css or styles).

Type

This attribute specifies the type of document being linked to. The value should be text/css.

rel

This specifies the relationship between the HTML page and the file it is linked to. The value should be stylesheet when linking to a CSS file. An HTML page can use more than one CSS style sheet. To do this it could have a element for every CSS file it uses. For example, some authors use one CSS file to control the presentation (such as fonts and colors) and a second to control the layout.

C.2 Link to External CSS

External style sheets can be referenced with a full URL or with a path relative to the current web page.

Example

This example links to a style sheet located in the html folder on the current web site:

```
<link rel="stylesheet" href="/html/styles.css">
```

Example

This example links to a style sheet located in the same folder as the current page:

```
<link rel="stylesheet" href="styles.css">
```

Which Style Sheet to Use

Generally, it is not very difficult to decide which style sheet to use (external, internal or inline). Simply keep the following advantages and disadvantages of each type of style sheet in your mind while making a choice.

Table: CSS style sheets

Style Sheet	Advantages	Disadvantages
External	It can be used to set the style of many documents.	It requires extra downtime to load the style sheet.
Internal	It can control the style of each document separately. No additional downtime is required to load the style sheet.	It needs to reapply style information for each document.
Inline	It can control the style of each element. Can be used to override any external or document style.	It needs to apply style information for each element. Need extra work. Documents are difficult to update.

C.4. Importance of using External Cascading Style Sheets

Here are a few reasons this is better.

- ✓ Easier Maintenance
- ✓ Reduced File Size
- ✓ Reduced Bandwidth
- ✓ Improved Flexibility

2.2. Test and validate the layout with reference to design specifications.

2.2.1 Element Selector

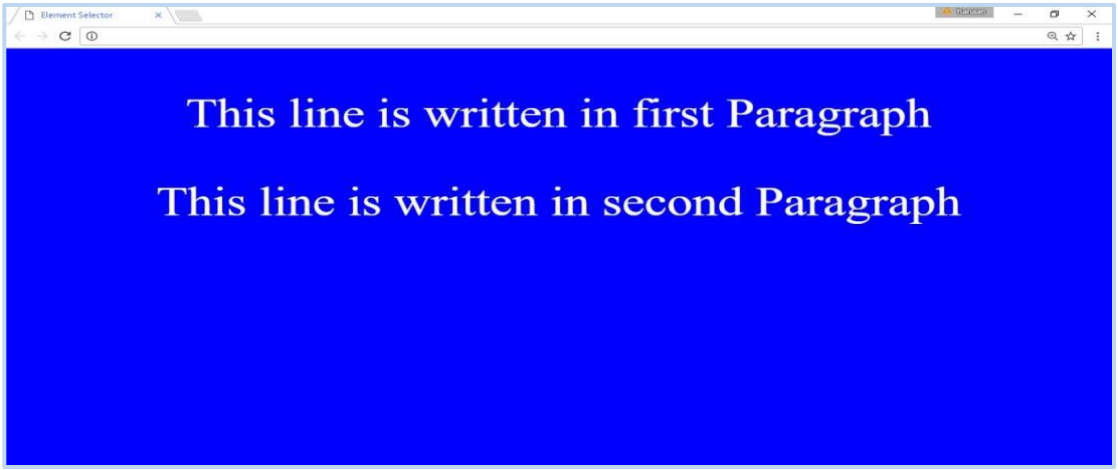
An element selector specifies the style for an element by name. For example, if we want to change the style of <p> element in a web page, we need to define a style for <p> as shown below.

```
p {   font-size: 25px;
       color: white;
        text-align: center;
      }
```

Once the style is defined, it will be applied to all paragraphs (tag <p>) automatically by the browser. A complete example of HTML code is given below. Note that we are using internal style sheets in our example. You can use external style sheets, if you like.

```
<!DOCTYPE html>
<html>
<head>
<title>Element Selector</title>
<style>
body   {
    background-color: blue;
  }    p   {     font-
size: 25px;     color:
white;      text-align:
center;
  }
</style>
</head>
<body>
  <p>This line is written in first Paragraph </p>
  <p>This line is written in second Paragraph </p>
</body>
 </html>
```

The output of above code is shown



2.2.2 ID Selector

The ID selector is used to select a specific HTML element. The ID of an element must be unique. To select an element with a specific ID, (#) character is used with ID name. Suppose we want to apply a style to one or more specific paragraphs (but not to all), then we should define the style as:

```
#p1
{
   font-size: 25px;
    color: white;
    text-align: center;
     }
```

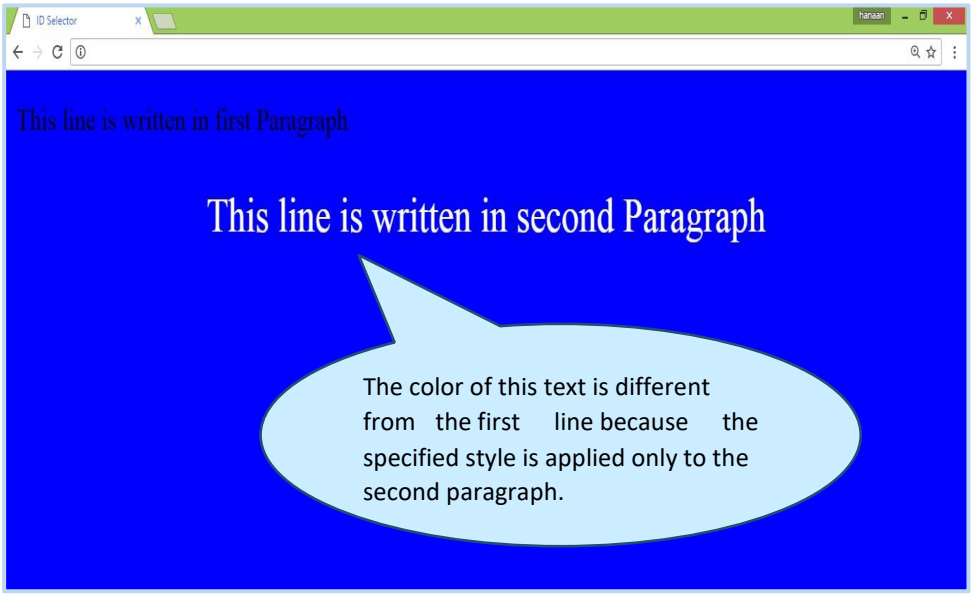
Now, we can use the above style with a paragraph by specifying the above style as shown below:

```
<p id="p1">This line is written in second Paragraph </p>
```

Note that the element selector applies to all whereas an ID selector is applied to the selected elements. We can define multiple styles for a single element. Try the following example and observe the output. We notice that style is applied to the second paragraph only.

```
<!DOCTYPE html>
<html>
<head>
<title>ID Selector</title>
<style>    body    {
      background-color: blue;
    }
    #p1     {
      font-size: 25px;       color: white;
text-align: center;
    }
</style>
</head>
<body>
   <p>This line is written in first Paragraph </p>
   <p id="p1">This line is written in second Paragraph </p>
</body>
</html>
```

The output of above code is



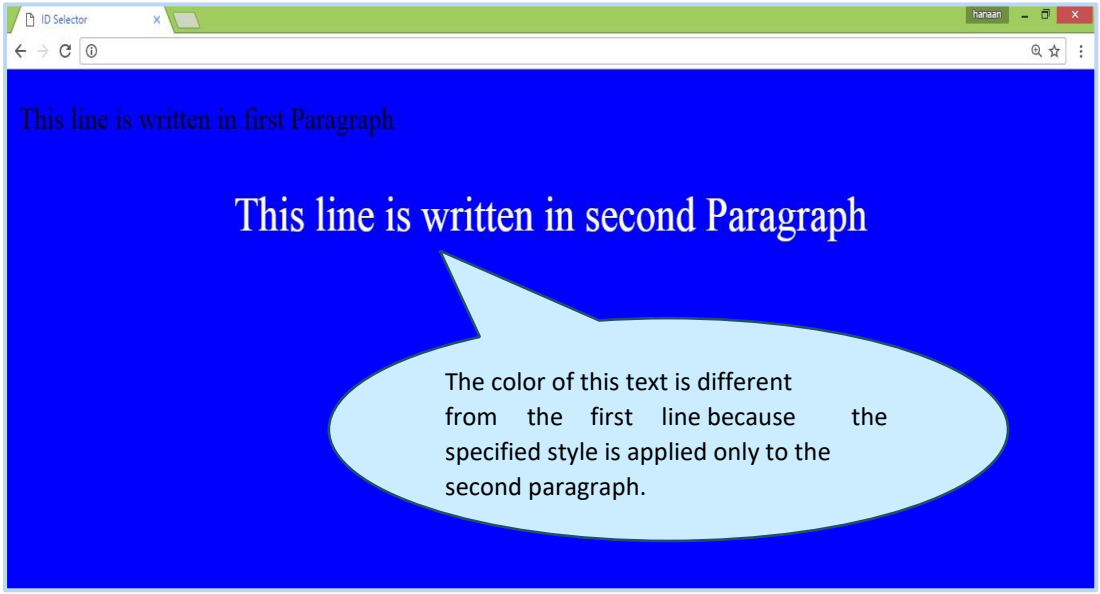
2.2.3 Class Selector

Class selector selects the element with a specific class attribute. To select an element of a specific class (.) character is used with class name. Suppose we have two paragraphs and want to apply different style on the

second paragraph. We give an ID to the second paragraph in order to apply different CSS properties. A class selector and an ID selector are somehow similar to each other, however, an ID selector identifies an element only, whereas a class selector can identify more than one elements.

```
<!DOCTYPE html>
<html>
<head>
<title>Class Selector</title>
<style>
body    {
      background-color: blue;
    }
   .p1
{
      font-size: 25px;
color: white;        text-
align: center;
    }
</style>
</head>
<body>
   <p>This line is written in first Paragraph </p>
   <p class="p1">This line is written in second Paragraph
</p> </body></html>
```

The output of above code is



2.2.4 CSS Comments

Writing a comment within a code is very useful because it helps to understand and modify the source code at later time. when you edit the source code at a later time, a comment is written as:

```
/* A comment */
```

The following code shows how useful comments can be written in an HTML document. See the code below how we can write useful comments in an HTML document.

```
<!DOCTYPE html>
<html>
<head>
<title>Element Selector</title>
<style>
body    {
     background-color: blue; /* Set background color to blue
*/    }
   .p1  /* Specify the style for paragraph */
   {
     font-size: 25px;
color: white;        text-align:
center;
    }
</style>
</head>
<body>
   <p>This line is written in first Paragraph </p>
   <p class="p1">This line is written in second Paragraph
</p>
</body>
</html>
```

2.2.5 Basic Properties of CSS

We discussed how rules are formed in CSS style sheets but did not mention what are various properties that can be set. CSS defines a large number of properties and their values.

Some of the basic properties are:

1. Background property
2. Border, margin & padding properties
3. Height / width property
4. Text
5. Fonts
6. links
7. Lists
8. Tables

We will discuss how to set the above properties in this and the next unit. Let us start with the background property.

2.2.5.1 Background Properties

Background properties are used to define the background effects for elements. Some of the background properties are given in the following table:

Background properties with their description

Property	Description
background-color	It specifies which background color is to be used
background-image	It specifies one or more background images to be used
background-repeat	It specifies how to repeat the background images
background-attachment	It specifies whether the background images are fixed or scrolls with the rest of the page
background-position	It specifies the position of the background images

Let us see how these properties are defined.

2.2.5.2 background-color

This property defines the background color of an HTML element or a document. The backgroundcolor property is defined as:

```
background-color: color;
```

A color can be specified in one of the following three methods:

1. Using a valid color name, like red
2. Using a RGB value, like rgb (255, 0, 0)
3. Using a hexadecimal value like #ff0000.

For example, to set background color of an HTML document, we specify the background-color property as:

```
bod
y {
    background-color: lightblue;
}
```

The following example shows how to set background color for a paragraph

```
p
{
    background-color: yellow;
}
```

Similarly, the following code can be used to set the background-color for <h1>.

```
h
1 {
```

```
    background-color: yellow;
}
```

See the HTML code below that sets the background color for the whole document to blue.

```
<!DOCTYPE html>
<html>
<head>
<title>Background Property in CSS</title>
<style>    body
{
    background-color: blue;
  }
</style>
</head>
<body>
  <h1>Playing with colors</h1>
  <p>This line is written in first Paragraph </p>
  <p>This line is written in second Paragraph </p>
</body>
</html>
```

2.2.5.3 background-image

We can use an image as a background of an HTML document to make it look like a wallpaper. The following example illustrates how to use background-image property:

```
bod
y {
      background-image: url("paper.gif");
}
```

By default, the image is repeated on the screen to cover the whole document. Try the following code and see the output in the browser.

```
<!DOCTYPE html>
<html>
<head>
<title>Element Selector</title>
<style>    body    {       background-image: url("new.jpg");     }
     p
  {
        color: white;        text-
  align: center;
     }
  </style>
  </head>
  <body>
     <p>This is Paragraph 1
  </p>
     <p>This is Paragraph
  2</p>
  </body>
  </html>
```

The output of above code is

2.2.5.4 Background repeat

Some images are designed in a way that they can be attached to each other, either horizontally or vertically, to form a certain design. By default, background-image property repeats the images both vertically and horizontally. To set the repeat direction of a background image, we use backgroundrepeat property that accepts one of two values: repeat-x or repeat-y. To prevent the image from being repeated, we use the value no-repeat.

The following example shows how to use background-repeat property. Keep in mind that background-repeat must be used along with background-image property and not by itself.

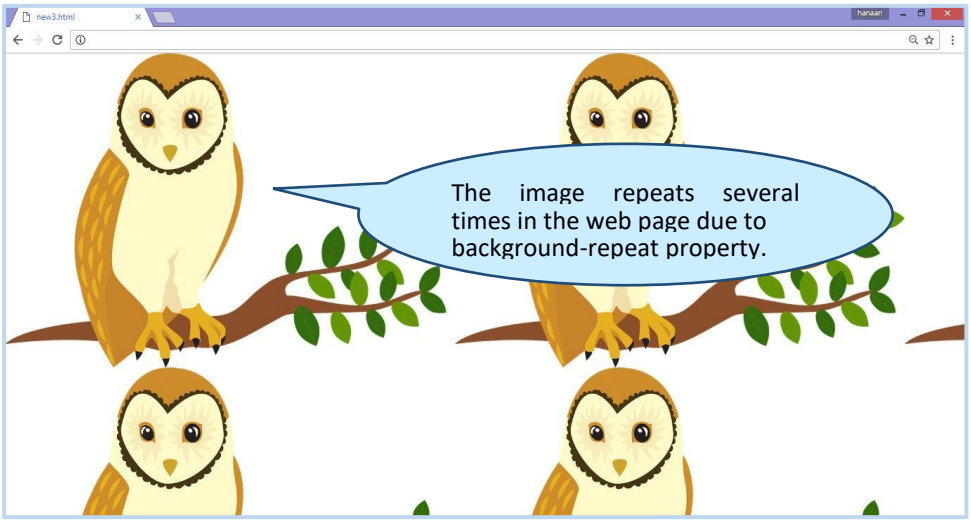
```
body  {     background-image: url("tree.png");
            background-repeat: repeat-x;
        }
```

The following code displays an image in the background with repeat property. The output is amazing.

```
<!DOCTYPE html>
<html>
<head>
<title>Element Selector</title>
<style>    body    {
      background-image: url("tree.png");
      background-repeat: repeat;
  }    p    {
      color: red;
       text-align: center;
  }
</style>
</head>
```

```
<body>
</body>
</html>
```

The output of above code is



2.2.5.5 background-attachment

This property determines whether that background image will scroll or not when the content with which it is associated scrolls. This can be specified by background-attachment property as shown in the following code.

```
bo
dy
{
    background-image: url("img_tree.png");
    background-repeat: no-repeat;
    background-attachment: fixed;
}
```

In the above code, we set the background-attachment to “fixed” which allow us to scroll up and down in the document without moving the background image.

2.2.5.6 background-position

This property determines the position of the background image within the canvas space used by its element. The background position can be set to "left top", "left center", "left bottom", "right top", "right center", "right bottom", "center top", "center center". If we specify only one value only, the second value is taken as center.

The following example defines the position of the background image as top right side of the document.

```
body
{
    background-image: url("tree.png");
    background-repeat: no-repeat;
    background-position: right top;
    background-attachment: fixed;
}
```

The following code displays an image on the top right side of the web page. We can see in the output that the image is displayed on the top right side of the web page, while “backgroundattachment” is used to fix the background image so that image will not scroll.

```
<!DOCTYPE html>
<html>
<head>
<title>Element Selector</title>
<style>   body     {
    background-color: blue;
    background-image: url("tree.png");
    background-repeat: no-repeat;
    background-position: right top;
  }   p    {      color:white;
                 text-align: center;
   }
</style>
</head>
<body>
    <p>This is Paragraph 1 </p>
    <p>This is Paragraph 2</p>
</body>
</html>
```

The output of code 5.12 is

Summary for the trainer related to the content (key notes using bullets such as ticks etc)

In this unit, we learned about advantages and the use of Cascading Style Sheet in web page development. We discussed how CSS code can be written to style different HTML elements. We have also discussed the differences and usage of external, internal and inline style sheets.

Theoretical learning Activity

Write short answers of the following questions.

1. What are the benefits of using CSS?
2. Differentiate between external, internal and inline style sheets?
3. How an external style sheet is linked to a HTML document?
4. What is CSS syntax to define properties of an HTML element?
5. List background properties and their possible values.

Practical learning Activity

1. Create the web page by removing the style from the HTML document and store it in a CSS file instead. Link your document with the style sheet.
2. Repeat the above activity by using inline style.

Points to Remember (Take home message)

- CSS treats each HTML element as if it appears inside its own box and uses rules to indicate how that element should look.
- Rules are made up of selectors (that specify the elements the rule applies to) and declarations (that indicate what these elements should look like).
- Different types of selectors allow you to target your rules at different elements.
- Declarations are made up of two parts: the properties of the element that you want to change, and the values of those properties. For example, the font-family property sets the choice of font, and the value arial specifies Arial as the preferred typeface.
- CSS rules usually appear in a separate document, although they may appear within an HTML page

- Use the HTML **style** attribute for inline styling
- Use the HTML **<style>** element to define internal CSS
- Use the HTML **<link>** element to refer to an external CSS file
- Use the HTML **<head>** element to store <style> and <link> elements

Learning outcome 2.1: formative assessment

Please mix different assessment instruments/tools for triangulation and relevancy of assessment

Written assessment

Choose the correct answer.

1. Which of the following is correct about CSS?
   A. CSS is used to control the style of a web document in a simple and easy way.

B. CSS is the acronym for "Cascading Style Sheet".
C. You can write CSS once and then reuse same sheet in multiple HTML pages.
D. All of the above.

2. Which of the following property is used to set the background image of an element?
   A. background-color
   B. background-image
   C. background-repeat
   D. background-position

3. The correct place to refer an external style sheet in an HTML document is
   A. <stylesheet>mystyle.css</stylesheet />
   B. <style src = "mystyle.css" />
   C. <link rel = "stylesheet" type = "text/css" href = "mystyle.css">
   D. None of the above

4. Where in an HTML document is the correct place to refer to an external style sheet?
   A. At the end of the document
   B. In the <head> section
   C. At the top of the document
   D. In the <body> section

5. Which HTML tag is used to define an internal style sheet?
   A. <style>
   B. <css>
   C. <script>
   D. All of the above

6. Which is the correct CSS syntax?
   A. body {color: black}
   B. {body;color:black}
   C. {body:color=black(body}
   D. body:color=black

7. How do you add a background color for all <h1> elements?
   A. all.h1 {background-color:#FFFFFF}
   B. h1.all {background-color:#FFFFFF}
   C. h1 {background-color:#FFFFFF}
   D. None of the above

8. What is the correct CSS syntax for making all the <p> elements bold?
   A. <p style = "text-size:bold">
   B. p {font-weight:bold}
   C. p {text-size:bold}
   D. <p style = "font-size:bold">

9. What is the correct CSS syntax for displaying background image?
   A. body { background-image: url("paper.gif"); }
   B. background.body: "paper.gif"
   C. body {background-image = "paper.gif"
   D. None of the above

10. Which CSS property is used to specify position of the background image?
   A. background-position
   B. image-position
   C. background-image-position
   D. None of the above

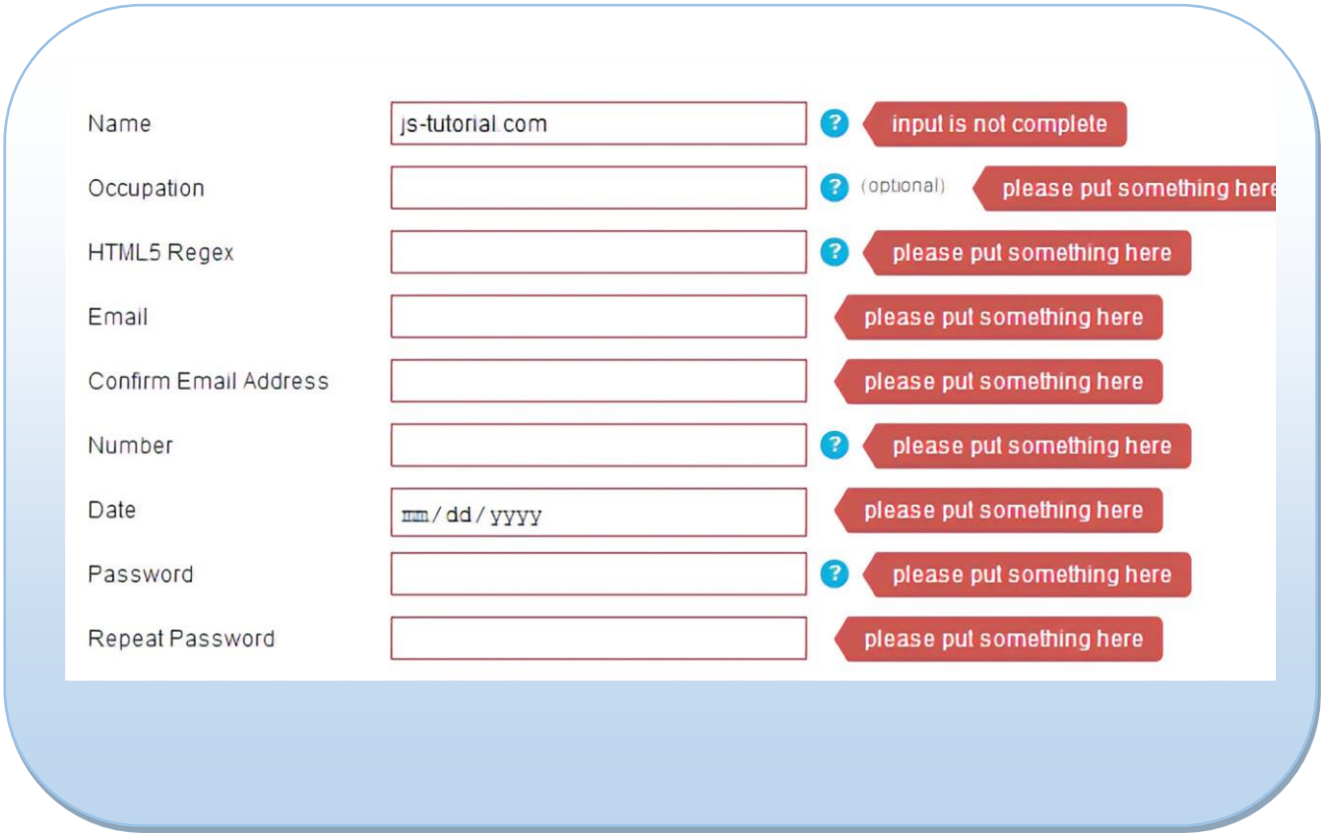
Answer Key:

1. D	2. B	3. C	4. B	5. A	6. A	7. C	8. B	9. A	10. A

Practical assessment

1. Create a CSS with suitable background properties for a web page that gives some basic information about your city (add some nice photos as well).
2. Create a website that gives basic information about yourself. You should prepare a separate page for each of the following: personal information, academic qualification, professional qualification, job experience, hobbies, and photo gallery. Academic qualification page should display the information about your degrees and certificates along with the subjects you studied in each in a tabular form. Create suitable CSS file and link it with all of your pages.

Learning Unit 3: Ensure website content meets technical protocols and standards.



STRUCTURE OF LEARNING UNIT

Learning outcomes:
3.1. Verify and validate the web content in accordance with required standards.
3.2. Test the content as per standard procedures.

Duration: 2hrs

Learning outcome 3.1 objectives:

By the end of Verify and validate the web content in accordance with required standards, the trainee will be able to:

- Required Field Validation
- Minimum Length Validation
- Maximum Length Validation
- Email Validation
- Number Validation
- URL Validation

Resources

Equipment	Tools	Materials
✓ Computer.	✓ Papers. ✓ Markers. ✓ Board.	✓ Books. ✓ Pens. ✓ Pencils.

CONTENT

Learning Outcome 3.1 Verify and validate the web content in accordance with required standards.

3.1.0 HTML Forms

Introduction

HTML Form is one of the main points of interaction between a user and a website or application. The user uses it to send data to the website. An HTML Form consists of one or more widgets. These widgets can be text fields (single line or multiline), select boxes, buttons, checkboxes, or radio buttons. Mostly, these widgets are paired with a label that describes their purpose.

An **HTML form** is *a section of a document* which contains controls such as text fields, password fields, checkboxes, radio buttons, submit button, menus etc.

An HTML form facilitates the user to enter data that is to be sent to the server for processing such as name, email address, password, phone number, etc.

An HTML form is used to collect user input. The user input is most often sent to a server for processing.

Example:

First name:
John
Last name:
Doe

Submit

Why use HTML Form

HTML forms are required if you want to collect some data from of the site visitor.

For example: If a user wants to purchase some items on internet, he/she must fill the form such as shipping address and credit/debit card details so that item can be sent to the given address.

HTML FORMS ELEMENTS

1. **The <form> Element**

The HTML **<form>** element provide a document section to take input from user. It provides various interactive controls for submitting information to web server such as text field, text area, password field, etc.

```
<form>
.
form elements
.
</form>
```

The HTML **<form>** element can contain one or more of the following form elements:

1. **<label>:** Defines a label for an <input> element
2. **<input>:** Defines an input control
3. **<select>:** Defines a drop-down list
4. **<textarea>:** Defines a multiligne input control (text area)
5. **<button>:** Defines a clickable button
6. **<fieldset>:** Groups related elements in a form
7. **<legend>:** Defines a caption for a <fieldset> element
8. **<output>:** Defines the result of a calculation
9. **<option>:** Defines an option in a drop-down list
10. **<datalist>**
11. **<optgroup>**

2. **The <label> Element**

The **<label>** element is useful for screen-reader users, because the screen-reader will read out loud the label when the user focus on the input element.

The **<label>** element represents a label which can be associated to a **<form>** control, and is supposed to provide a short description for it.

Browsers may link both elements by allowing users to set the focus to the control by clicking on its label.

Example:

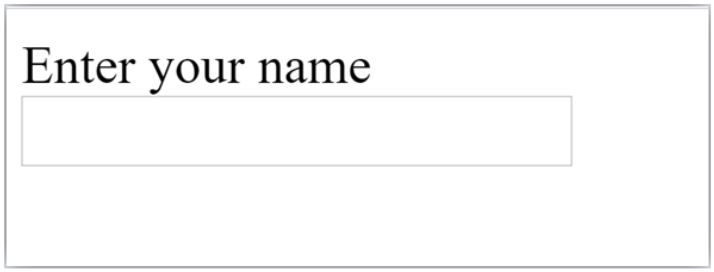
```
<form > <p><label>Username: <input type="text" name="user"></label></p>
</form>
```

3. The <input> Element

The HTML **<input>** element is fundamental form element. It is used to create form fields, to take input from user. We can apply different input filed to gather different information form user. Following is the example to show the simple text input.

```
<body>
  <form>
    Enter your name <br>
   <input type="text">
  </form>
</body>
```

Output:



Here are the different input types you can use in HTML:

- <input type="text">
- <input type="password">
- <input type="button">

- <input type="checkbox">
- <input type="color">
- <input type="date">
- <input type="datetime-local">
- <input type="email">
- <input type="file">
- <input type="hidden">
- <input type="image">
- <input type="month">
- <input type="number">
- <input type="radio">
- <input type="range">
- <input type="reset">
- <input type="search">
- <input type="submit">
- <input type="tel">
- <input type="time">
- <input type="url">
- <input type="week"

1. <input type="text">:

<input> element of type "**text**" is used to define a single-line input text field.

Example:

```
<form>
  <label>Enter first name</label><br>

  <input type="text" ><br>
  <label>Enter last name</label><br>

  <input type="text"><br>
</form>
```

Output:

Enter first name

Enter last name

2. <input type="password">:

The <input> element of type "**password**" allow a user to enter the password securely in a webpage. The entered text in password filed converted into "*" or ".", so that it cannot be read by another user.

Example:

```
<form>
    <label>Enter User name</label><br>
    <input type="text"><br>
    <label>Enter Password</label><br>
    <input type="Password"><br>
</form>
```

Output:

Enter User name

RTB

Enter Password

••••••

3. <input type="submit">:

The <input> element of type "**submit**" defines a submit button to submit the form to the server when the "click" event occurs.

Example:

```
<form action=" ">
   <label>Enter User name</label><br>
   <input type="text"><br>
  <label> Enter Password</label><br>
   <input type="Password"><br>
   <br><input type="submit" >
</form>
```

Example:

Enter User name
RTB
Enter Password
•••••••

Submit

4. <input type="reset">:

The <input> type "**reset**" is also defined as a button but when the user performs a click event, it by default reset the all inputted values.

Example:

```
<form action=" ">
   <label>Enter User name</label><br>
   <input type="text"><br>
   <label>Enter Password</label><br>
   <input type="Password"><br>
   <br><input type="submit" >
<br><input type="reset" >
</form>
```

Output:

Enter User name

RTB

Enter Password

••••••

Submit Reset

5. <input type="radio">:

The <input> type "**radio**" defines the radio buttons, which allow choosing an option between a set of related options. At a time only one radio button option can be selected at a time.

Example:

```
<form>
  <p>Kindly Select your favorite color</p>

  <input type="radio"  value="red"> Red <br>
  <input type="radio"  value="blue"> blue <br>

  <input type="radio"  value="green">green <br>
  <input type="radio"  value="pink">pink <br>

  <input type="submit" value="submit">
</form>
```

Output:

Kindly Select your favorite color

○ Red

◉ blue

○ green

pink

submit

Bottom of Form

6. <input type="checkbox">:

The <input> type "**checkbox**" are displayed as square boxes which can be checked or unchecked to select the choices from the given options.

Example:

```
<form>
<h2>Registration Form</h2>
    <label>Enter your Name:</label>

    <input type="text" name="name">
    <p>Kindly Select your favourite sports</p>

    <input type="checkbox" name="sport1" value="cricket">Cricket<br>
    <input type="checkbox" name="sport2" value="tennis">Tennis<br>

    <input type="checkbox" name="sport3" value="football">Football<br>
    <input type="checkbox" name="sport4" value="baseball">Baseball<br>


    <input type="checkbox" name="sport5" value="badminton">Badminton
<br>
    <br>
    <input type="submit" value="submit">

  </form>
```

Output:

Registration Form

Enter your Name:

Kindly Select your favorite sports

☐ Cricket

☐ Tennis

☐ Football

☐ Baseball

☐ Badminton

submit

Bottom of Form

N.B: The "**radio**" buttons are similar to **checkboxes**, but there is an important difference between both types: radio buttons allow the user to select only one option at a time, whereas checkbox allows a user to select zero to multiple options at a time**.**

7. <input type="button">:

The <input> type "**button**" defines a simple push button, which can be programmed to control a functionally on any event such as, click event.

Example:

```
<form>
    <input type="button" value="Clcik me ">
</form>
```

8. <input type="file">:

The <input> element with type "**file**" is used to select one or more files from user device storage. Once you select the file, and after submission, this file can be uploaded to the server

Example:

```
<form>
    <label>Select file to upload:</label>
    <input type="file" name="newfile">
    <input type="submit" value="submit">
</form>
```

9.<input type="image">:

The <input> type "**image**" is used to represent a submit button in the form of image.

Example:

```
<!DOCTYPE html>
<html>
<body>
<h2>Input "image" type. </h2>
<p>We can create an image as submit button</p>
 <form>
   <label>User id:</label><br>
    <input type="text" name="name"><br><br>
    <input type="image" alt="Submit" src="login.png"  width="100px">
 </form>
</body>
</html>
```

10.<input type=" Color ">:

The <input type="color"> is used for input fields that should contain a color.

Depending on browser support, a color picker can show up in the input field.

Example

```
<form>
  <label for="favcolor">Select your favorite color:</label>
  <input type="color"  name="favcolor">
</form>
```

11.<input type="date">:

The <input type="date"> is used for input fields that should contain a date.

Depending on browser support, a date picker can show up in the input field.

Example

```
<form>
  <label for="birthday">Birthday:</label>
  <input type="date"  name="birthday">
</form>
```

You can also use the **min** and **max attributes** to add restrictions to dates:

Example

```
<form>
  <label for="datemax">Enter a date before 1980-01-01:</label>
  <input type="date"  name="datemax" max="1979-12 31"><br><br>
  <label for="datemin">Enter a date after 2000-01-01:</label>
  <input type="date"  name="datemin" min="2000-01-02">
</form>
```

12.<input type="email">:

The <input type="**email**"> is used for input fields that should contain an e-mail address.

Depending on browser support, the e-mail address can be automatically validated when submitted.

Some smartphones recognize the email type, and add ".com" to the keyboard to match email input.

Example

```
<form>
  <label for="email">Enter your email:</label>
  <input type="email"  name="email">
</form>
```

13.<input type="month">

The <input type="**month**"> allows the user to select a month and year.

Depending on browser support, a date picker can show up in the input field.

Example

```
<form>
  <label >Birthday (month and year):</label>
  <input type="month"  name="bdaymonth">
</form>
```

14.<input type="number">

The <input type="**number**"> defines a **numeric** input field.

You can also set restrictions on what numbers are accepted.

The following example displays a numeric input field, where you can enter a value from 1 to 5:

Example

```
<form>
  <label>Quantity (between 1 and 5):</label>
  <input type="number"  name="quantity" min="1" max="5">
</form>
```

Example

```
<form>
  <label>Quantity:</label>
  <input type="number"  name="quantity" min="0" max="100" step="10" v
alue="30">
</form>
```

15.<input type="range">:

The <input type="**range**"> defines a control for entering a number whose exact value is not important (like a slider control). Default range is 0 to 100. However, you can set restrictions on what numbers are accepted with the **min**, **max**, and **step** attributes:

Example

```
<form>
  <label >Volume (between 0 and 50):</label>
  <input type="range"  name="vol" min="0" max="50">
</form>
```

16.<input type="search">

The <input type="**search**"> is used for search fields (a search field behaves like a regular text field).

Example

```
<form>
  <label for="gsearch">Search Google:</label>
  <input type="search" name="gsearch">
</form>
```

17.<input type="tel">:

The <input type="**tel**"> is used for input fields that should contain a telephone number.

Example

```
<form>
  <label for="phone">Enter your phone number:</label>
  <input type="tel" name="phone" pattern="[0-9]{3}-[0-9]{2}-[0-9]{3}">
</form>
```

18.input type="time">

The <input type="**time**"> allows the user to select a time (no time zone).

Depending on browser support, a time picker can show up in the input field.

Example

```
<form>
  <label for="appt">Select a time:</label>
  <input type="time" name="appt">
</form>
```

19.<input type="url">

The <input type="**url**"> is used for input fields that should contain a URL address.

Depending on browser support, the url field can be automatically validated when submitted.

Some smartphones recognize the **url type**, and adds ".com" to the keyboard to match url input.

Example

```
<form>
  <label for="homepage">Add your homepage:</label>
  <input type="url"  name="homepage">
</form>
```

20.<input type="week">

The <input type="**week**"> allows the user to select a week and year.

Depending on browser support, a date picker can show up in the input field.

Example

```
<form>
  <label for="week">Select a week:</label>
  <input type="week"  name="week">
</form>
```

3.The <select> Element

The **<select>** element defines a drop-down list:

Example

```
<label for="cars">Choose a car:</label>
<select name="cars">
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
```

```
  <option value="audi">Audi</option>
</select>
```

Output:

Choose a car: Volvo

The **<option>** elements define an option that can be selected.

By default, the first item in the drop-down list is selected.

To define a pre-selected option, add the **selected attribute** to the option:

Example

```
<option value="fiat" selected>Fiat</option>
```

Visible Values:

Use the **size attribute** to specify the number of visible values:

Example

```
<label for="cars">Choose a car:</label>
<select  name="cars" size="3">
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

Allow Multiple Selections:

Use the **multiple attribute** to allow the user to select more than one value:

Example

```
<label for="cars">Choose a car:</label>
<select  name="cars" size="4" multiple>
  <option value="volvo">Volvo</option>
  <option value="saab">Saab</option>
  <option value="fiat">Fiat</option>
  <option value="audi">Audi</option>
</select>
```

4.The <textarea> Element

The **<textarea>** element defines a multi-line input field (a text area):

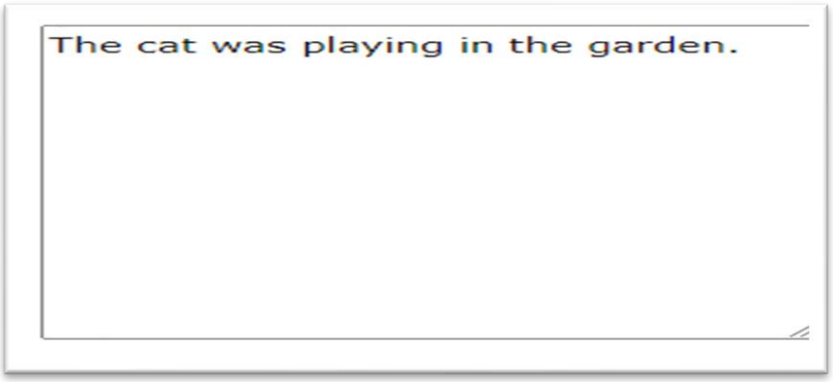
Example

```
<textarea name="message" rows="10" cols="30">
The cat was playing in the garden.
</textarea>
```

The **rows attribute** specifies the visible number of lines in a text area.

The **cols attribute** specifies the visible width of a text area.

This is how the HTML code above will be displayed in a browser:



You can also define the size of the text area by using CSS:

Example

```
<textarea name="message" style="width:200px; height:600px;">
The cat was playing in the garden.
</textarea>
```

5.The <button> Element

The **<button> element** defines a clickable button:

Example

```
<button type="button" >Click Me!</button>
```

This is how the HTML code above will be displayed in a browser:

Click Me!

6.The <fieldset> and <legend> Elements

The **<fieldset>** element is used to group related data in a form.

The **<legend>** element defines a caption for the **<fieldset>** element.

Example

```
<form action="/action_page.php">
  <fieldset>
    <legend>Personalia:</legend>
    <label for="fname">First name:</label><br>
    <input type="text"  name="fname" value="John"><br>
    <label for="lname">Last name:</label><br>
    <input type="text"  name="lname" value="Doe"><br><br>
    <input type="submit" value="Submit">
  </fieldset>
</form>
```

This is how the HTML code above will be displayed in a browser:

Personalia:
First name:
John
Last name:
Doe
Submit

7.The <datalist> Element

The **<datalist>** element specifies a list of pre-defined options for an **<input>** element.

Users will see a drop-down list of the pre-defined options as they input data.

The **list attribute** of the **<input> element**, must refer to the id attribute of the **<datalist>** element.

Example

```
<form action="/action_page.php">
  <input list="browsers">
  <datalist id="browsers">
    <option value="Internet Explorer">
    <option value="Firefox">
    <option value="Chrome">
    <option value="Opera">
    <option value="Safari">
  </datalist>
</form>
```

8.The <output> Element

The **<output>** element represents the result of a calculation (like one performed by a script).

Example

Perform a calculation and show the result in an **<output>** element:

```
<form action="/action_page.php"
  oninput="x.value=parseInt(a.value)+parseInt(b.value)">
  0
  <input type="range"  id="a" name="a" value="50">
  100 +
  <input type="number" id="b" name="b" value="50">
  =
  <output name="x" for="a b"></output>
  <br><br>
  <input type="submit">
</form>
```

HTML Form Attributes

This chapter describes the different attributes for the HTML **<form>** element.

1. **The Action Attribute**

The **action attribute** defines the action to be performed when the form is submitted.

Usually, the form data is sent to a file on the server when the user clicks on the submit button.

In the example below, the form data is sent to a file called "action_page.php". This file contains a server-side script that handles the form data:

Example

On submit, send form data to "action_page.php":

```
<form action="/action_page.php">
  <label for="fname">First name:</label><br>
```

```
  <input type="text"  name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text"  name="lname" value="Doe"><br><br>
  <input type="submit" value="Submit">
</form>
```

If the **action attribute** is omitted, the action is set to the current page.

2. The Method Attribute

The **method attribute** specifies the HTTP method to be used when submitting the form data.

The form-data can be sent as URL variables (with **method="get"**) or as HTTP post transaction (with **method="post"**).

The default HTTP method when submitting form data is **GET**.

Example

This example uses the **GET method** when submitting the form data:

```
<form  action="/action_page.php" method="get">
```

Example

This example uses the **POST method** when submitting the form data:

```
<form action="/action_page.php" method="post">
```

Difference between GET and POST:

GET:

- Appends the form data to the URL, in name/value pairs
- NEVER use GET to send sensitive data! (The submitted form data is visible in the URL!)
- The length of a URL is limited (2048 characters)

- Useful for form submissions where a user wants to bookmark the result
- GET is good for non-secure data, like query strings in Google

POST:

- Appends the form data inside the body of the HTTP request (the submitted form data is not shown in the URL)
- POST has no size limitations, and can be used to send large amounts of data.
- Form submissions with POST cannot be bookmarked

3. The Target Attribute

The **target attribute** specifies where to display the response that is received after submitting the form.

The **target attribute** can have one of the following values:

Value	Description
_blank	The response is displayed in a new window or tab
_self	The response is displayed in the current window
_parent	The response is displayed in the parent frame
_top	The response is displayed in the full body of the window
framename	The response is displayed in a named iframe

The default value is **_self** which means that the response will open in the current window.

Example

Here, the submitted result will open in a new browser tab:

```
<form action="/action_page.php" target="_blank">
```

4. **The value Attribute**

The input **value attribute** specifies an initial value for an input field:

Example

Input fields with initial (default) values:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text"  name="fname" value="John"><br>
  <label for="lname">Last name:</label><br>
  <input type="text"  name="lname" value="Doe">
</form>
```

5. **The readonly Attribute**

The input **readonly attribute** specifies that an input field is read-only.

A read-only input field cannot be modified (however, a user can tab to it, highlight it, and copy the text from it).

The value of a read-only input field will be sent when submitting the form!

Example

A read-only input field:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text"  name="fname" value="John" readonly><br>
  <label for="lname">Last name:</label><br>
  <input type="text"  name="lname" value="Doe">
</form>
```

6. The disabled Attribute

The input **disabled attribute** specifies that an input field should be disabled.

A disabled input field is unusable and un-clickable.

The value of a disabled input field will not be sent when submitting the form!

Example

A disabled input field:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text"  name="fname" value="John" disabled><br>
  <label for="lname">Last name:</label><br>
  <input type="text" name="lname" value="Doe">
</form>
```

7. The size Attribute

The input **size** attribute specifies the visible width, in characters, of an input field.

The default value for size is 20.

Note: The size attribute works with the following input types: text, search, tel, url, email, and password.

Example

Set a width for an input field:

```
<form>
  <label for="fname">First name:</label><br>
```

```
  <input type="text"  name="fname" size="50"><br>
  <label for="pin">PIN:</label><br>
  <input type="text"  name="pin" size="4">
</form>
```

8. The maxlength Attribute

The input **maxlength** attribute specifies the maximum number of characters allowed in an input field.

N.B: When a **maxlength** is set, the input field will not accept more than the specified number of characters. However, this attribute does not provide any feedback.

Example

Set a maximum length for an input field:

```
<form>
  <label for="fname">First name:</label><br>
  <input type="text"  name="fname" size="50"><br>
  <label for="pin">PIN:</label><br>
  <input type="text"  name="pin" maxlength="4" size="4">
</form>
```

9. The min and max Attributes

The input **min** and **max** attributes specify the minimum and maximum values for an input field.

The **min** and **max** attributes work with the following input types: number, range, date, datetime-local, month, time and week.

Example

Set a max date, a min date, and a range of legal values:

```
<form>
  <label for="datemax">Enter a date before 1980-01-01:</label>
  <input type="date"  name="datemax" max="1979-12-31"><br><br>

  <label for="datemin">Enter a date after 2000-01-01:</label>
  <input type="date"  name="datemin" min="2000-01-02"><br><br>

  <label for="quantity">Quantity (between 1 and 5):</label>
  <input type="number"  name="quantity" min="1" max="5">
</form>
```

10. The multiple Attribute

The input **multiple** attribute specifies that the user is allowed to enter more than one value in an input field.

The **multiple** attribute works with the following input types: email, and file.

Example

A file upload field that accepts multiple values:

```
<form>
  <label for="files">Select files:</label>
  <input type="file"  name="files" multiple>
</form>
```

11. The pattern Attribute

The input **pattern** attribute specifies a regular expression that the input field's value is checked against, when the form is submitted.

The **pattern** attribute works with the following input types: text, date, search, url, tel, email, and password.

Example

An input field that can contain only three letters (no numbers or special characters):

```
<form>
  <label for="country_code">Country code:</label>
  <input type="text"  name="country_code"
  pattern="[A-Za-z]{3}" >
</form>
```

12. The placeholder Attribute

The input **placeholder** attribute specifies a short hint that describes the expected value of an input field (a sample value or a short description of the expected format).

The short hint is displayed in the input field before the user enters a value.

The **placeholder** attribute works with the following input types: text, search, url, tel, email, and password.

Example

An input field with a placeholder text:

```
<form>
  <label for="phone">Enter a phone number:</label>
  <input type="tel"  name="phone"  placeholder="123-45-678"
  pattern="[0-9]{3}-[0-9]{2}-[0-9]{3}">
</form>
```

Enter a phone number: 123-45-678

13. The required Attribute

The input **required** attribute specifies that an input field must be filled out before submitting the form.

The **required** attribute works with the following input types: text, search, url, tel, email, password, date pickers, number, checkbox, radio, and file.

Example

A required input field:

```
<form>
  <label for="username">Username:</label>
  <input type="text"  name="username" required>
</form>
```

14. The height and width Attributes

The input **height** and **width** attributes specify the height and width of an <**input type="image"**> element.

Example

Define an image as the submit button, with height and width attributes:

```
<form>
  <label for="fname">First name:</label>
  <input type="text"  name="fname"><br><br>
  <label for="lname">Last name:</label>
  <input type="text"  name="lname"><br><br>
  <input type="image" src="img_submit.gif" alt="Submit" width="48" height="4
8">
</form>
```

Learning Outcome3.2 Test the content as per standard procedures

Form Validation

Form validation is a "*technical process where a web-form checks if the information provided by a user is correct.*"

The form will either alert the user that they messed up and need to fix something to proceed, or the form will be validated and the user will be able to continue with their registration process.

Example:

Twitter won't let me use an email address that is incorrectly formatted, and gives me an error message.

Types of form validation.

In general, there are main types of form validation:

- Required Field Validation
- Minimum Length Validation
- Maximum Length Validation
- Email Validation
- Number Validation
- URL Validation
- Pattern Validation or Regular Expression Validation

1. **Required Field Validation:**

   If you want to make an input mandatory to be entered by the user, you can use the required attribute. This attribute can be used with any input type such as email, URL, text, file, password, checkbox, radio, etc. This can help to make any input field mandatory.

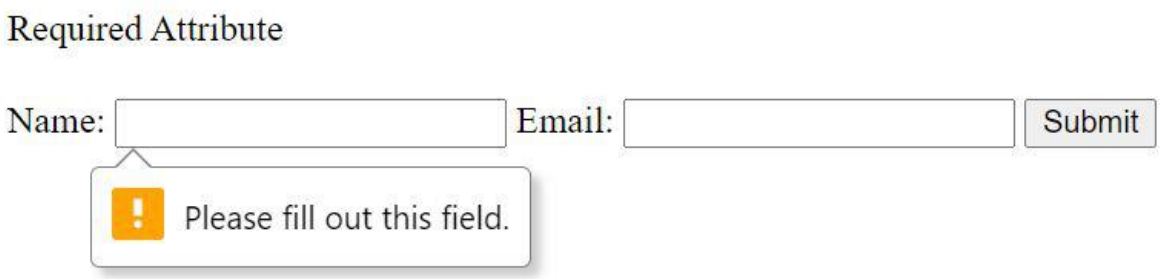
Syntax:

<input type="..." id="..." name="..." **required**>

Example:

```
<!DOCTYPE html>
<html lang="en">
<body>
   <p>Required Attribute</p>
   <form>
      <label>Name:</label>
      <input type="text" required />
      <label>Email:</label>
      <input type="email" required />
      <button style="margin-top: 5px">Submit</button>
   </form>
</body>
 </html>
```

Output:



2. **Maximum Length Validation**

The maxlength attribute helps us to specify the **maximum length** of an input field in the form i.e. the maximum no. of characters that can be entered

for the specific field. It can be used to make the maximum length of a telephone number in India as 10 digits so as to prevent any wrong input data submission.

Syntax:

<input type="..." id="..." name="..." **maxlength**="...">

N.B: When the maxlength attribute is set, it will not allow the user to enter more characters than those specified.

Example:

```
<!DOCTYPE html>
<html lang="en">
<body>
    <p>Maxlength Attribute</p>
    <form>
        <label>Roll No.</label>
        <input type="text" maxlength="10" required />
        <label>Email:</label>
        <input type="email" required />
        <button style="margin-top: 5px">Submit</button>
    </form>
</body>
</html>
```

Output:

Maxlength Attribute

Roll No. 3674357435 Email: Submit

3. Minimum Length Validation

The minlength attribute helps us to specify the **minimum length** of an input field in the form i.e., the minimum no. of characters that must be entered for the specific field. It can be useful to make the minimum length of a password that can't be guessed or a telephone number in India as 10 digits so as to prevent any wrong input data submission.

Syntax:

```
<input type="..." id="..." name="..." minlength="...">
```

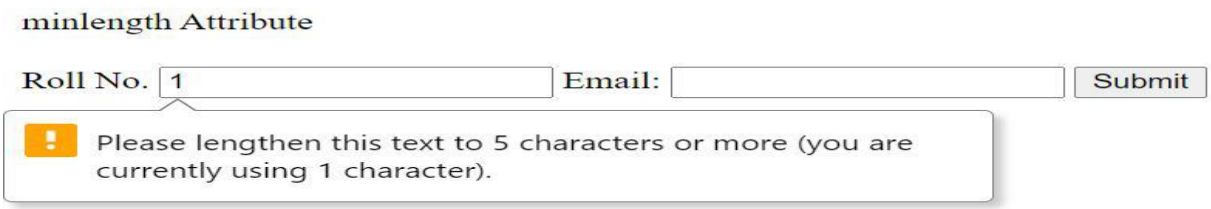
N.B: When the minlength attribute is set, an input field will not accept lesser characters than those specified.

Example:

```
<!DOCTYPE html>
<html lang="en">
<body>
  <p>minlength Attribute</p>
   <form>
     <label>Roll No.</label>
     <input type="text" minlength="5" required />
     <label>Email:</label>
     <input type="email" required />
     <button style="margin-top: 5px">Submit</button>
  </form>
```

```
</body>

</html>
```

Output:



4. Min and max attributes:

The min and **max attributes** of the input field are used to specify the **minimum** and **maximum** allowed values for an input field. We can create a range of allowed values using min and max together.
min specifies the minimum value an input field can have whereas max specifies the maximum value for a field.

The min and max attributes work with the various input types such as number, range, date, datetime, month, time and week. We can use these attributes in a field where we want a user to enter some amount. We can set the maximum limit and minimum limit for the amount to be entered.

Syntax:

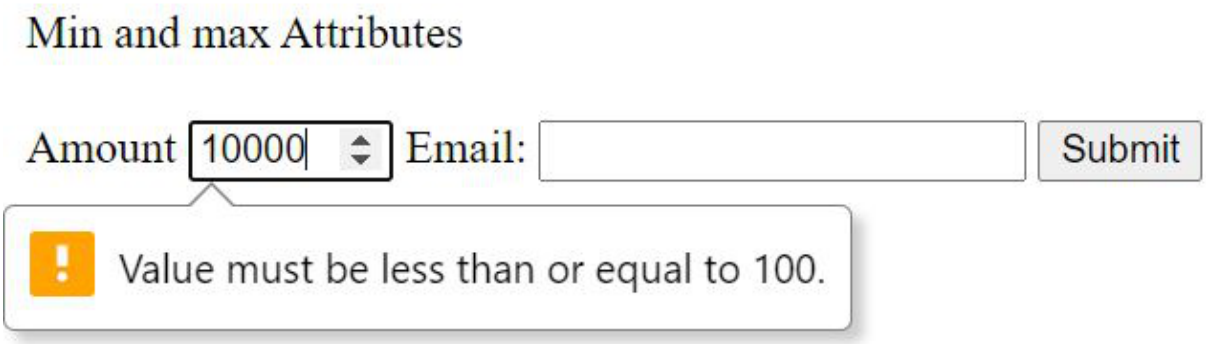
```
<input type="..." id="..." name="..." max="...">

<input type="..." id="..." name="..." min="..." max="...">
```

Example:

```
<!DOCTYPE html>

<html lang="en">

<body>

    <p>Min and max Attributes</p>

    <form>
```

```
        <label>Amount</label>
        <input type="number" min="50" max="100" required />
        <label>Email:</label>
        <input type="email" required />
        <button style="margin-top: 5px">Submit</button>
    </form>
</body>
</html>
```

Output:



5. Pattern Validation:

The **pattern attribute** defines a particular pattern that an input field's value must satisfy when the form is submitted. The pattern attribute works with

text, date, search, URL, tel, email, and password input types. We make use of regex or **regular expressions** to specify the pattern. It is useful in situations where we want the input entered must be according to the given pattern only. The inputs that do not satisfy the pattern will not be accepted by the form.

Syntax:

```
<input type="...." id="..." name="...." pattern=".....">
```

Example:

```
<!DOCTYPE html>
<html lang="en">
<body>
    <p>Pattern Attribute</p>
    <form>
        <label>Tel.</label>
        <input type="text" pattern="[0-9]{10}" required />
        <label>Email:</label>
        <input type="email" multiple required />
        <button style="margin-top: 5px">Submit</button>
    </form>
</body>
</html>
```

Output:

Pattern Attribute

Tel. hdjdj Email: Submit

Please match the requested format.

In the above example, the pattern specifies that Tel. can be composed of digits only and its length must be 10.

6. **Email Validation**

To validate an email address of an input field, just set the input type to "**Email**".

Example

```
<form name="Form">

  <input type="Email" name="Email" ng-model="user.Email"  />

</form>
```

7. **Number Validation**

To validate that an input field has a number, just set the input type to "**number**".

Example

```
<form name="Form">

  <input type="number" name="number" ng-model="user.Age" />

</form>
```

8. **URL Validation**

To validate that an input field has an URL, set the input type to "URL".

```
<form name="Form">

   <input type="url" name="homepage" ng-model="user.google_url" />

</form>
```

Theoretical learning Activity

Write short answers of the following questions.

1. What is difference between checkboxes and radio buttons?
2. Discuss various input types available for a form.

Practical learning Activity

1. Create a web page that gives some basic information about your city (add some nice photos as well). At the end of the page, create a simple form for more information form that allows the users to input their names, email addresses, and comments.
    a. Add a checkbox on the web page asking the reader either he likes to join the group.
    b. Add a drop down list that allows the user to select his/her country.

Points to Remember (Take home message)

Form: Allows a user to enter data on a web page.

Textbox: It is a rectangular area for entering text in a web page.

Dropdown list: It is a list of items for selecting a value from pull down menu.

Radio button: It is used to select one option from set of many option

Check box: It is used to select more than one options.

Formative assessment

Please mix different assessment instruments/tools for triangulation and relevancy of assessment

Written assessment

Choose the correct answer.

1. Which of the following tag is used to specify a multi-line text area in a form?
   A. <input type="Multi-text">
   B. <textarea> <textarea>
   C. <form type="textarea">
   D. None of the above

2. A form can be placed anywhere in the head section of a HTML document.
   A. True
   B. False

3. Which of the following code is used to create a single-line text box in a form?
   A. <input type="text" name="name">
   B. <input type="text-box" name="name">
   C. <form type="text" name="name">
   D. <input type="box" name="name">

4. <input type = “submit”> creates a button on a form.
   A. True
   B. False

5. Which of the following code clears all the form elements?
   A. <input type="clear">
   B. <form type="reset">
   C. <input type="reset">
   D. <input type="reset-form">

6. Which of the following code is used to create a radio button?
   A. <input type="radio-button" name = "gender" value = "male">
   B. <input type="radio" name = "gender" value = "male">
   C. <input type="button" name = "gender" value = "male" >
   D. None of the above

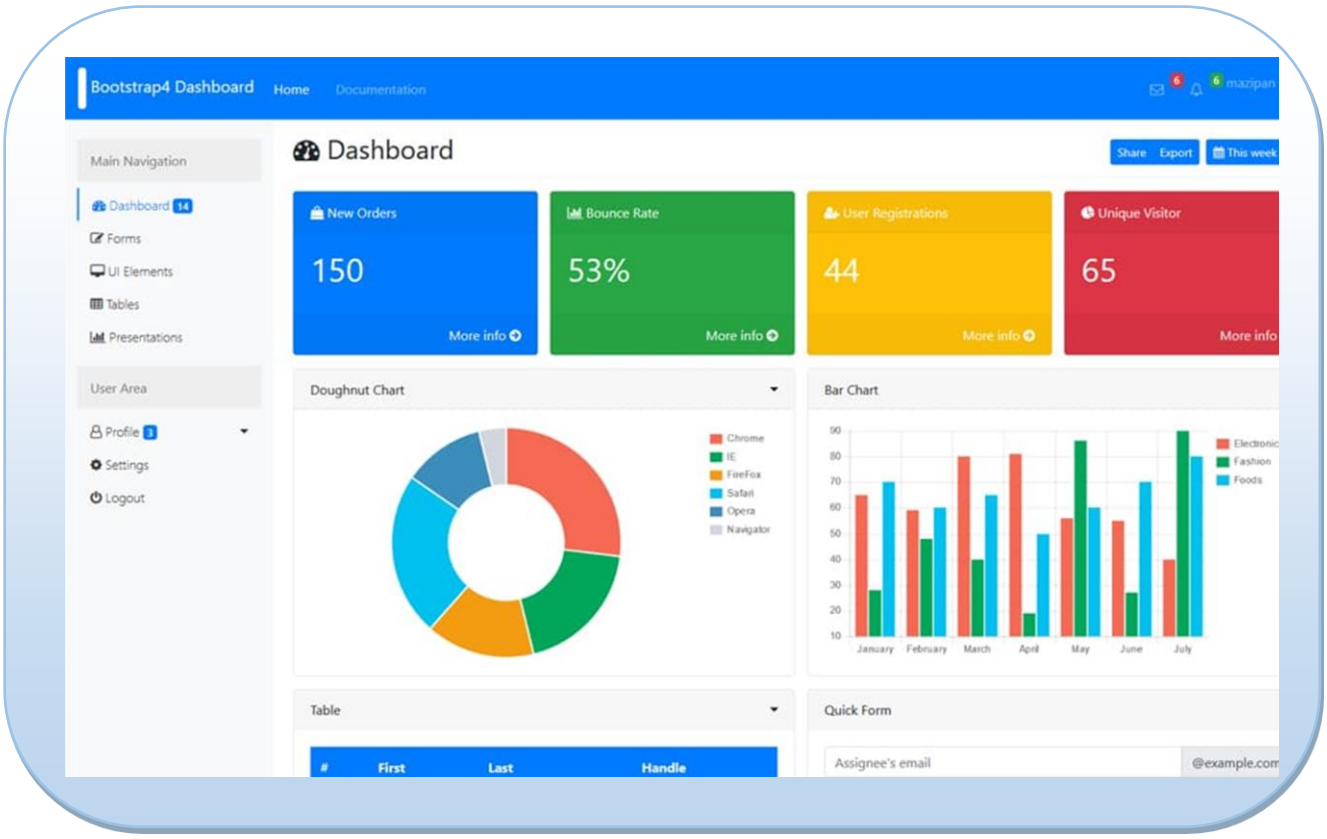
7. <input type="checkbox" name="choice" value="Orange"> will create a check box in a form.
   A. True
   B. False

8. Which input type is used to create display a calendar?
   A. Date
   B. Calendar
   C. Date-Time
   D. None of the above
9. Which of the following tag is used to specify items in a select-box?
   A. <choice>
   B. <item>
   C. <select>
   D. Option

Answer Key:

1. B	2. B	3. A	4. A	5. C	6. B	7. A	8. A	9. C

Learning Unit 4: Produce interactive web animations



STRUCTURE OF LEARNING UNIT

Learning outcomes:

4.1. Analysis of website requirements in accordance with client's requirements

4.2. Design of web animations according to the design specifications

4.3. Publish animation

Duration: 2hrs

Learning outcome objectives :

By the end of Produce interactive web animations, the trainee will be able to:

- Analysis of website requirements in accordance with client's requirements
- Design of web animations according to the design specifications
- Publish animation

Resources

Equipment	Tools	Materials
✓ Computer.	✓ Papers. ✓ Markers. ✓ Board.	✓ Books. ✓ Pens. ✓ Pencils.

CONTENT

Learning Outcome 4.1: Analyze website requirements in accordance with client's requirements

So, you've just signed your terms and engaged in website design services with Primal Space, but what next? We're often asked by clients before a website design project commences; 'What is required?' Therefore, we've decided to write everything down in this handy guide making life much easier for you, our new web design client, going forward. Below we have detailed exactly what you will need. Helping you prepare to get started with the project, before work at our end can commence.

No matter what stage you are at with your website design idea. You can use this blog article as a great reference point. It's worth noting that this is a general guide we have written, not every concept in this article may apply to your website design. But the majority certainly will, as most of the list are core aspects of any website. However, it's important to note that if you are having your website designed by our good selves, the information that we detail here will be required before we start building and designing your website.

If you aren't yet at that stage, you might want to go back a step and check out the post about gathering specifications and the goals of your new website. We've also created a new more in depth post titled how to create requirements documents which includes example templates that you can follow.

Website Design Requirements

Your idea and vision for the project is arguably the most important part of the whole preparation process. Tell us about your idea! What purpose do you want the website to serve?

It's important to have everything in a clear structure that is easy for us to follow. Your ideas and visions can be collated into documents or mood boards if you like and provided as such to us by scanning and uploading to our Google Drive.

Or we can meet over video call or face to face for coffee and you can tell us all about your new website concept.

Colour Scheme

If you have a particular colour (or colour spectrum) that needs to be featured throughout the website. You'll need to specify this to us. We'll require the 6-digit hexadecimal colour code for the colour you want to use. We recommend using the Hex colour picker website here Hex Colour Picker. For a website design, we generally like to stick to a base of 3 or 4 colours at most. With 2 of those colours generally being black or white. If you've read other articles on our website you'll already know that we are lovers of white space within a website design. Allowing your content to take centre stage so to speak! The colour you provide to us from the hex colour chart will generally be your brand colours. Or the main colour that you want to be used through the theme of your website design and other marketing material including your logo.

Fonts

Have you decided on specific fonts that you'd like to be used within the design to depict your brands image? Just like colour, these fonts should be consistent throughout all marketing materials as well as company logos to create a concise brand identity.

Website Content Requirements

Forming the nuts and bolts of your website so to speak is your websites content! We've broken this concept up into a list of images, mock-ups, text wording content and your logo. Depending on your website requirements what you will need may differ, or you may need to provide us with more content. For example these could include, videos, PDF or any other graphics that you require within the design. Your mock-up provided should reference the file names of any particular content you require within the website design.

Images

We'll let you into a little secret, eye-catching images that give an instant wow factor are an integral part of any website design. If you have really strong and powerful images, the design of your website will instantly look more impressive. We would thoroughly recommend hiring a professional photographer where possible. Providing us with the most high-resolution photos you can get your hands on. It will really make a difference to the final design and help your website stand out alongside your competitors.

If your budget can't stretch to paying for a professional photographer, then perhaps looking at how you can obtain copyright free imagery for your website online would be beneficial. There are lots of great websites out there where you can access high quality photos free or for a donation.

Mock-ups

A mock-up in terms of website design is a document showing how you want that page of the website to be laid out. We receive mock-ups from clients in many forms. Some are more sophisticated than others. However it's an integral part of the preparation process for your website design. Without a mock-up being provided to us, it's very difficult to get a clear idea of how you want your content displayed on each page. There are a whole variety of formats in which a website design mock-up can be provided to us. You can provide hand-drawn mock-ups to us. Or you can go to the other end of the spectrum, providing professional wire-frames showing each page of your website. Mock-ups can be provided as a Word Document, or PDF. Or like we say if you are hand drawing your mock-up, you can scan these and upload them to Google Drive.

You may also wish to provide us with website screenshots representing particular elements or functions of existing websites that you would like to be featured within your own website.

Text Content

The wording for text content should be detailed in a separate document for each page. The text content can be provided to us as a Word document (.doc) or as a Text document (.txt), either is fine.

Before your project commences we'll have established how you'll provide your content to us via a shared access Google Drive folder. This is all part of your website design project management plan which we will have put in place and provided you with ahead of development. This is a requirement of all Primal Space digital servicees.

Logo

If your logo is not being purchased through us, you'll need to provide us with this. The logo asset files should be provided as a .ai or .eps file type. Allowing us to edit the logo's design and size where required to fit within your websites pages. The logo is often edited purely for the purpose of the website design.

If you don't yet have a logo for your website, reach out to us for a quote for our graphic design service. Allowing us to design a professional logo for your business as part of the build.

Access to your Domain and Hosting

If your domain was not purchased through us here at Primal Space. And/or you have your hosting with another company. For us to be able to implement your website live, we'll require login access to our domain and/or hosting control panel to apply the settings. Even if you have your domain purchased elsewhere. But you are having your website hosting supplied through ourselves Primal Space. We'll still need access to the domain settings to apply changes there to point to the new website.

Learning Outcome 4.2 Design of web animations and responsiveness according to the design specifications

An Introduction to Animation in Web Design

Animation is not just for cartoons anymore. From full-screen moving images to small hover effects, touches of animation are popping up everywhere. Animation is trendy, fun, and user-friendly.

Animation happens when something created in still or two-dimensional form is "brought to life" and appears to move in a way that follows laws of physics. It's the way a cartoon character walks across the screen or how an app icon bounces like a ball while it is loading on the desktop of your Mac.

One of the words that is almost synonymous with animation is Disney. In the early 1980s, two of the companies' top animators wrote a book detailing the 12 principles of animation. The "**Illusion of Life: Disney Animation**" by Frank Thomas and Ollie Johnston still provides the framework for animation today.

1. Squash and stretch
2. Anticipation
3. Staging
4. Straight Ahead Action and Pose to Pose
5. Follow Through and Overlapping Action
6. Slow In and Slow Out
7. Arc
8. Secondary Action
9. Timing
10. Exaggeration
11. Solid Drawing
12. Appeal

Web animations are often saved as GIF, CSS, SVG, WebGL or video. They can be anything from a simple underline that appears when you hover over a word to a full-screen video or background image. As with any other design technique, animations can be subtle or they might be in your face and hard to avoid.

To help guide your animation decisions, you might want to avoid some more trendy elements and stick to some of the more timeless elements of animation design. These techniques provide a basic guide that can help you create something usable and visually interesting.

- **Navigation:** Use micro-animations to show users when they are activating an element or menu item ahead of the click or tap
- **Hover states:** Another micro-animation that can facilitate user experience with a cue to what is actionable
- **Video:** This major animation can be an engaging content type that helps tell a story with the design
- **Sliding transitions:** Made popular in hero headers, this method moves images or illustrations through a scroll or automatic action (the trick here is to make each transition as smooth as possible)
- **Scrolling:** From parallax effects to more complex animated transitions, the scroll is a great place to add subtle motion to denote shifts in content
- **Background animations:** Subtle elements of motion can make a static image more interesting and provide an element of surprise for users
- **Welcome or loading animations:** Many welcome page animations hide a load issue, but when done properly the motion is interesting enough that it's not annoying and distracts the user from the lag

- **Storytelling animation:** Motion can help set the scene for the content therein and can help keep users on a website longer as they get into the tale at hand

The primary reason to use animation is to increase usability. Simple animations can be great guiding tools to help people understand what buttons to click or where to go next in the map of a website. Many designers using complex scrolling effects pair a simple animation with a user tool to scroll or click. (This includes everything from a simple bouncing icon or words that pop up and say "scroll down.")

Usability comes in a few forms:

- Communication function or how to use a website
- Attract users to the overall design or a specific element
- Increase interaction and engagement
- Show change, such as filling in a form correctly or highlighting that an element is clickable
- Establish a better or stronger emotional connection
- Create flow or direct users to a call-to-action
- Improve the overall user experience

Animation can also be a great "decoration." Sometimes the goal of an animated element is purely visual and that is an acceptable use. This decorative animation can help tell a story or create an emotional connection between the interface and the user. The purpose of an animation can be to spark visual interest and keep a user engaged with your site for a longer period of time.

When creating a purely visual animation consider what it is supposed to do. Think about the connection you want a user to have. Is it supposed to be fun or surprising? Is it a bit of unique content that is designed for sharing? Even a pure visual should have a goal.

Learning Outcome 4.3: Publish animations

Animations in an acceptable web format

web animations into a hypertext markup language (HTML) page

Straight forward and simple web pages are good but not the best. The animation is the thing that brings life to a static page and makes it more eye-catchy.

Here are a few ideas of animation for your web pages using the frontend languages like HTML & CSS.

1. Rotating a Card: The first one is rotating a card. The card has two things front side and the backside when we hover our mouse to the card it will rotate 180 degrees and it will show the backside only.

Explanation: Here the first whole body is styled the image. The button is styled according to your need and creativity. The main thing here is the rotation at first the backside is hidden or rotated 180 degrees and when we hover to the card the front side is rotating -180 degrees and the backside is coming in the front.

Below is the example that illustrates the *Rotating card* animation.

Example:

```
<!DOCTYPE html>
<html lang="en">

<head>
        <style>
                body {
```

```
        color: hsl(209, 61%, 16%);
        background: #fff;
        ;
}
img {
        width: 100%;
        display: block;
}
.btn {
        background: white;
        color: rgb(64, 119, 64);
        padding: 0.375rem 0.75rem;
        letter-spacing: 3px;
        display: inline-block;
        border: 2px solid transparent;
        border-radius: .5rem;
        cursor: pointer;
}
.blog {
        background: hsl(210, 36%, 96%);
}
.card {
        height: 500px;
        position: relative;
        width: 500px;
        margin-left: 30rem;
}
.card-side {
```

```
        -webkit-transition: all 2s linear;
        transition: all 2s linear;
        -webkit-backface-visibility: hidden;
        backface-visibility: hidden;
        position: absolute;
        top: 0;
        left: 0;
        width: 100%;
        height: 100%;
        border-radius: 0.5rem;
        visibility: visible;
}
.card-back {
        background: rgb(55, 116, 75);
        -webkit-transform: rotateY(180deg);
        transform: rotateY(180deg);
        display: grid;
        place-items: center;
}
.card:hover .card-front {
        -webkit-transform: rotateY(-180deg);
        transform: rotateY(-180deg);
}
.card:hover .card-back {
        -webkit-transform: rotateY(0);
        transform: rotateY(0);
}
```

```
            .card-info {
                    padding: 1rem 1.5rem;
            }
            .card-front img {
                    height: 13rem;
                    -o-object-fit: cover;
                    object-fit: cover;
                    height: 500px;
            }
        </style>
</head>
<body>
        <div class="section blog">
                <div class="card">
                        <!-- front of the card -->
                    <div class="card-side card-front">
                        <img src=
"https://media.geeksforgeeks.org/wp-content/uploads/20201113113041/paintorder1.png"
                                alt="" />
                    </div>
                    <!-- card back -->
                    <div class="card-side card-back">
                        <button class="btn">
                                Read More
                        </button>
                    </div>
                </div>
        </div>
```

```
</body>
</html>
```

Output:

2. Text Moving Up & Down: In this animation, the text will go up on hovering the mouse to it, and in the case of mobile on click, the text will go up.

Explanation: At first, the down texts are being covered using the overflow is hidden and bottom negative value. And when we are hovering into the card the front text is going up using translate negative values and down text is shown by making the bottom value as zero. Linear-gradient is used to read the text clearly. Transition plays a big part as it makes things going smoothly. The rest of the code is just for styling and for making things good.

3. Text Coming up on Hovering: In this animation text coming up on hovering to the image and the image is zoomed.

Explanation: The first is to make the text disappear which is done by making opacity zero. Now on hovering the image the image will be zoomed and the overflow hidden property makes sure that the outer part of the image when it is zoomed doesn't go outside of the certain height and width. And on hovering

making the opacity of the image text one makes it visible. Transition property just made things smoother.

Bootstrap

Bootstrap is a free and open-source front end development framework for the creation of websites and web apps. The Bootstrap framework is built on HTML, CSS, and JavaScript (JS) to facilitate the development of responsive, mobile-first sites and apps.

Responsive design makes it possible for a web page or app to detect the visitor's screen size and orientation and automatically adapt the display accordingly; the mobile first approach assumes that smartphones, tablets and task-specific Mobile apps are employees' primary tools for getting work done and addresses the requirements of those technologies in design.

Bootstrap includes user interface components, layouts and JS tools along with the framework for implementation. The software is available precompiled or as source code.

In the physical world, a bootstrap is a small strap or loop at the back of a leather boot that enables you to pull the entire boot on and in general usage, bootstrapping is the leveraging of a small initial effort into something larger and more significant. There is also a common expression, "pulling yourself up by your own bootstraps," meaning to leverage yourself to success from a small beginning.

Appealing Features of Bootstrap

Bootstrap has a lot of features. These features not only make it stand out, but they also make it more popular even among those web designers who like to take things in a very conventional way.

i. Easy to Begin With

It is pretty easy, to begin with. Being easy to get started with is probably the first quality which makes Bootstrap very appealing.

b. LESS as Well as CSS Files

Bootstrap not only offers LESS files but also includes the old CSS files.

c. Easily Customizable

Despite the fact that Bootstrap is designed in responsive 12-column grids, layouts, and components, it is also very easy to customize. Whether you need a fixed grid or a responsive one, it can be made possible by making a few changes. Offsetting and nesting of columns are also easy to do in both CPU-based and mobile-based browser grids.

B. Responsive Utility Classes

Another prominent feature of Bootstrap is its responsive utility classes. Using responsive utility classes, a particular piece of content can be made to appear or hide only on devices depending on the size of the screen being used. This feature is extremely helpful for designers who want to make a mobile and tablet-friendly version of their websites.

C. Components of Bootstrap

Some of the components that come pre-styled in Bootstrap are:

- Drop-downs
- Button
- Navigation

- Badges Alerts
- Progress Bar

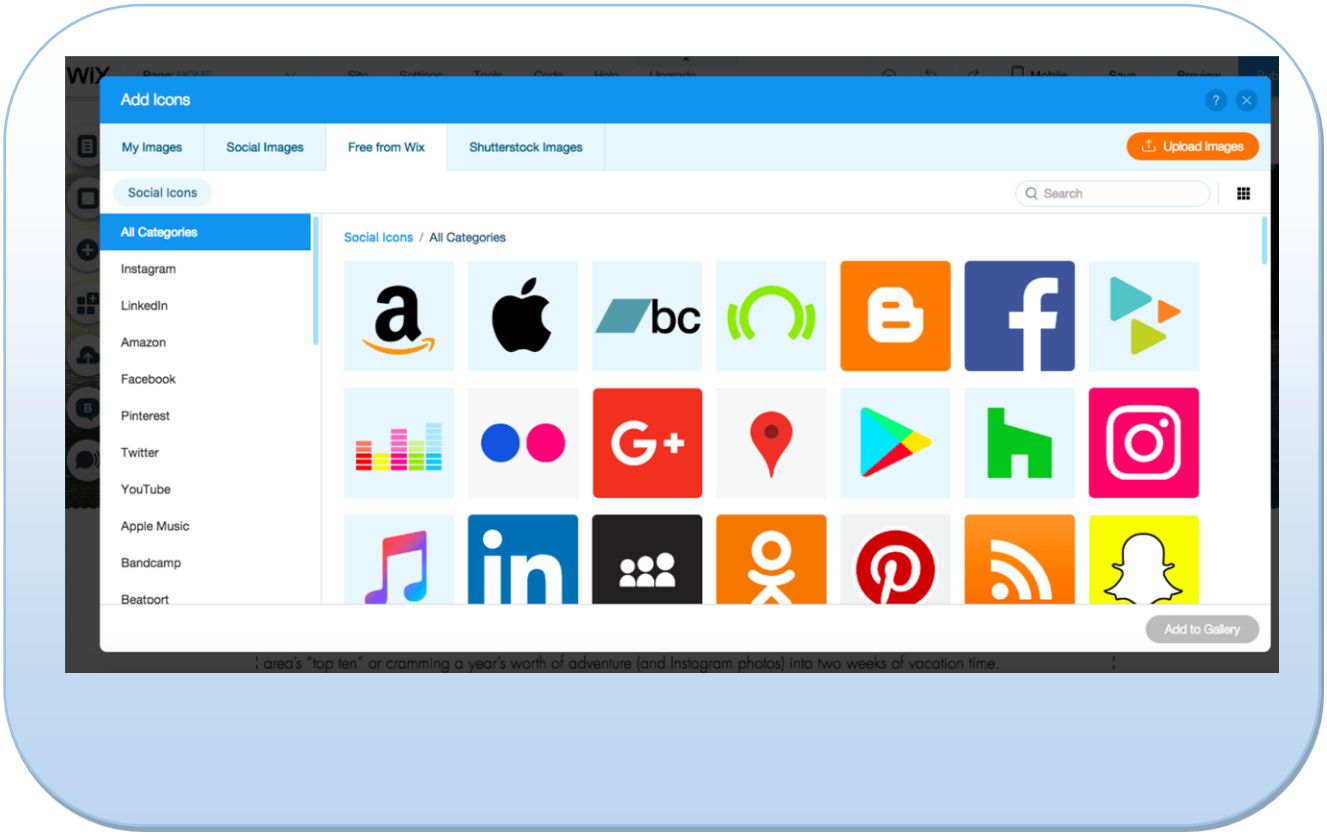
D. Drop-Down Component Menu

The drop-down component menu is a responsive additive feature of a website. To include it in a website, a lot of different plugins, mostly Java-based, are tested. But, via Bootstrap and its easy customizing options, this can easily be done in a matter of minutes.

E. Bootstrap Templates

The readily available templates make it easier for inexperienced users to create a website following a simple tutorial or demo available on the Bootstrap.

Learning Unit 5: Integrate website with social web technologies



STRUCTURE OF LEARNING UNIT

Learning outcomes:
5.1. Identify social web to be integrated with website
5.2. Develop the code and documentation of the integration requirements

Duration: 2hrs

Learning outcome objectives :

By the end of Integrate website with social web technologies, the trainee will be able to:

- Identify social web to be integrated with website
- Develop the code and documentation of the integration requirements

Resources

Equipment	Tools	Materials
✓ Computer.	✓ Papers. ✓ Markers. ✓ Board.	✓ Books. ✓ Pens. ✓ Pencils.

CONTENT

Learning Outcomes: 5.1 Identify social web to be integrated with website

Deffinition of social media

The term social media refers to a computer-based technology that facilitates the sharing of ideas, thoughts, and information through virtual networks and communities. Social media is internet-based and gives users quick electronic communication of content, such as personal information, documents, videos, and photos. Users engage with social media via a computer, tablet, or smartphone via web-based software or applications

Types of social media

Social media may take the form of a variety of tech-enabled activities. These activities include photo sharing, blogging, social gaming, social networks, video sharing, business networks, virtual worlds, reviews, and much more.

Integrate social media into Your Website

Having your website and social media accounts set up is just the beginning; you need to make sure that these different platforms are **effectively integrated with one another** so that customers are able to interact and engage with your brand regardless of time, place, and purpose.

The process of cross-promoting your social media accounts on your website so these communication channels can be more frequently used, explored, and shared is called **social media integration**. This is something that your business needs to do if you are going to be competitive in today's marketplace. **The ones who did this well will reap the benefits, and the ones that don't…well, won't.**

Ways To Integrate social media Into Your Website

Employ the following tactics in order to increase traffic amongst platforms, strengthen brand awareness and affinity, encourage communication with customers, and improve sales and revenue.

1. Social Media Icons

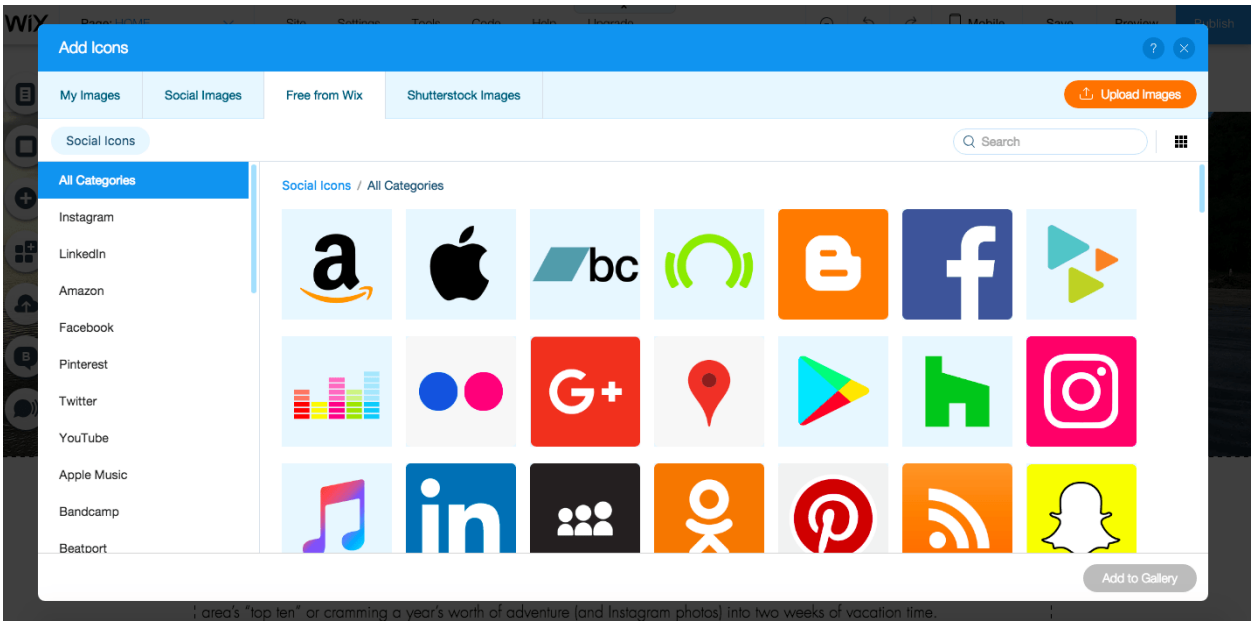
The first and the most obvious step on the road to integration includes **placing visible social media buttons on your website**, linking to your various profiles. Depending on your type of business and audiences, you might have a number of different social media profiles, such as:

- Facebook
- LinkedIn
- Twitter
- YouTube
- Pinterest

Naturally, you would like your website's visitors to also visit these profiles and consume the messages posted there. The best way to ensure this is to publish social media buttons on your website landing pages, this gives visitors have the ability to start following your accounts immediately, without even leaving your website.

Their placement on your website is very important; **there are certain locations that individuals are used to looking, so you'll want to be sure that their placement is intuitive**. Experts say the best places are either the top or bottom navigation bar (or both!) or visibly separated from additional content along the side of the homepage.

If you use a website development platform, social media icon integration is incredibly easy, and can be set up in less than a minute, which means that there's no excuse not to have these icons integrated across your website.



2. Social Share Buttons

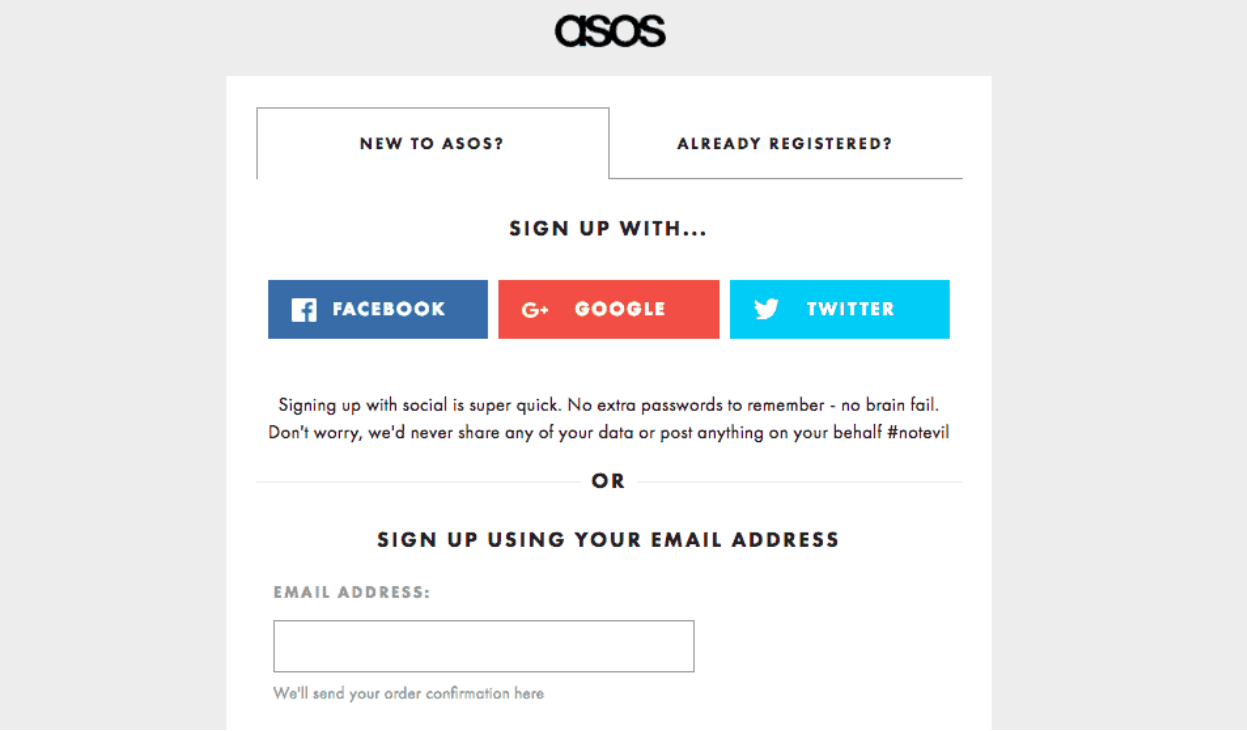
Regardless of whether your website is pushing out blog posts or featuring products, **you want your content to be shareable** so that visitors can easily post to their own social media accounts directly from your website. "**Share this**" or "**Add this**" are examples of share buttons across e-commerce websites with the possibility to see analytics of who and how your content is shared.

This not only increases the number of eyeballs on your content, but promotes your overall brand image and improves sale rates.

3. Social Login Plugins

Incorporate a social sign-in to allow your visitors to sign into your e-commerce website through one of their social media accounts. Research has shown that **users prefer to bypass the need to create a separate account** for a website and sign in with a social login instead.

Reducing the registration process **saves time for the customers, encourages users along the sales process, and enables brands to quickly capture customers' information** and engage them.



4. Social Live Feeds

Social live feeds allow you to display your live activity from Facebook, Twitter, Instagram, YouTube, or other social media accounts directly on your website. So, instead of merely linking to these social media channels, you can **show your current and recent social activity**, so your visitors can see, follow, and otherwise engage without leaving your website.

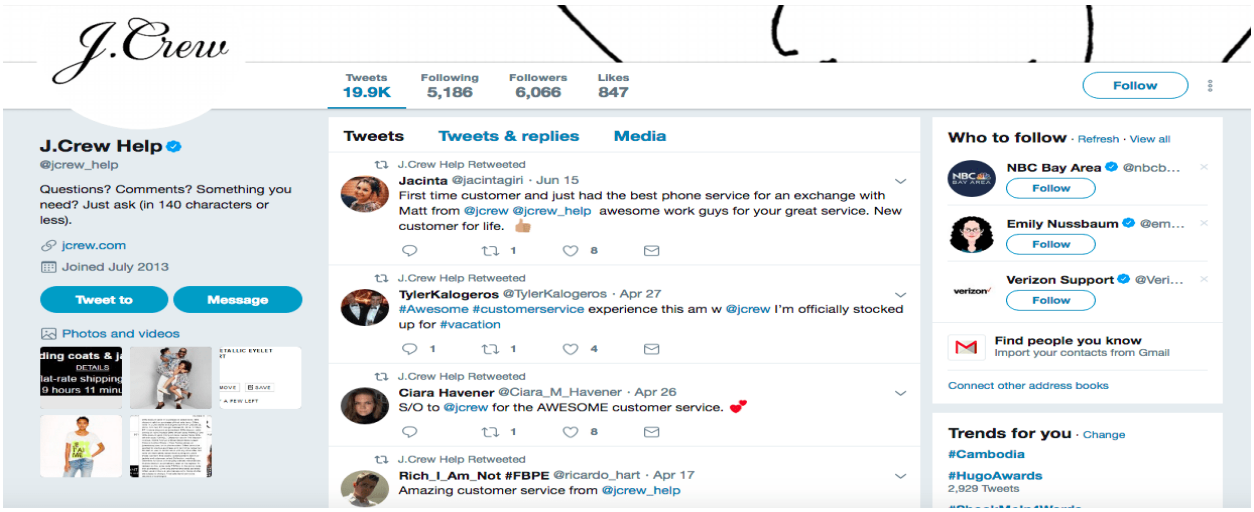
These feeds can make your website **feel less like a sales pitch and more human**. It also makes the entire brand experience more integrated, and increases the likelihood that someone will begin following your social media accounts while visiting your website.

Most social media platforms have widgets that allow you to easily customize settings, grab the code, paste and publish it directly onto a website. Depending on your industry, product or service offering, and audience, you may want to consider integrating your:

- Instagram feed
- Live Twitter stream
- YouTube recent videos
- Blog posts

5. Social Assistance Account

Customers not only appreciate, but often expect, immediacy. It has been shown that brands that are able to offer immediate assistance not only have **higher brand affinity, but customers are more likely and willing to pay a higher price** for their products and/or services.



Learning Outcome 5.2. Develop the code and documentation of the integration requirements

An *Integration Requirements Document* assesses and describes the requirements for a successful data integration. Much like a Systems/Software Requirements Specification (SRS), the Integration Requirements articulates the expected behavior and features of the integration project and related systems. Most Integrations Requirements Documents are part of a larger Data Integration Requirement Plan and quality of service standards. Typically, integration requirements documents include timelines, goals, expectations, rules, and roles as well as company specific requirements that will encompass complete data integration from all contributors of the process.

Documents carefully outline expectations and show a homogenous agreement among the company as to how to move forward with integration.

Integration Requirement Documents include scope and input documents. They often start with establishing a current evaluation of the business and data system. Then, they will explain the problem the integration is solving for and the business operations affected. As part of this, an assessment of the current data landscape is performed.

The Data Integration Requirements Documents also provide a detailed model for the kind of integration project and architecture that will be performed, something like a blueprint. The data integration model will define what each of the architectural details are and their function. For example, will it be a cloud-based data lake or an in-house data warehouse?

The Data Integration Requirements will include all software, hardware, features, and processes that are included in the data integration.

10 integration requirements

Currently, most application integration is done through REST and SOAP services, which have been determined to meet the majority of the needs on the integration requirements document. Integration requirements have changed, particularly as data management and use has been shaped by newer, cloud-based utility.

SnapLogic has outlined 10 of the requirements for the most contemporary, highest-functioning integrations. The "10 Modern Data Integration Platform Requirements" for a unified iPaaS are:

1. Application integration is done through SOAP and REST services
2. Large capacity for data integration and storage through a data lake or cloud-based data warehouse
3. Integration will be able to support the present and coming data velocities whether batch or streaming data
4. Event-based integration over clock-driven integrations
5. A document-centric data integration strategy
6. Hybrid integration, including for both cloud-to-cloud and cloud-to-ground scenarios.
7. Integration must be accessible via REST/SOAP APIs
8. Instead of location, new integrations must be focused on connectivity for speed and agility
9. Elastic integration able to operate in real-time and fluctuating event-based demands
10. SaaS-based approach wherein integration is delivered as a service to anyone who needs acces

Remember (Take home message)

- ✓ Types of social media
- ✓ Ways To Integrate social media Into Your Website
- ✓ Social media integration strategies for your website

Formative assessment

Please mix different assessment instruments/tools for triangulation and relevancy of assessment

Written assessment

1. How do I integrate social media into website?
2. Why you should integrate social media platforms into website?

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